Finite-Control-Set Model Predictive Control of Permanent Magnet Synchronous Motor Drive Systems — An Overview

Teng Li, Xiaodong Sun, *Senior Member, IEEE*, Gang Lei, *Member, IEEE*, Zebin Yang, Youguang Guo, *Senior Member, IEEE*, and Jianguo Zhu, *Senior Member, IEEE*

***Abstract*—Permanent magnet synchronous motors (PMSMs) have been widely employed in the industry. Finite-control-set model predictive control (FCS-MPC), as an advanced control scheme, has been developed and applied to improve the performance and efficiency of the holistic PMSM drive systems. Based on the three elements of model predictive control, this paper provides an overview of the superiority of the FCS-MPC control scheme and its shortcomings in current applications. The problems of parameter mismatch, computational burden, and unfixed switching frequency are summarized. Moreover, other performance improvement schemes, such as the multi-vector application strategy, delay compensation scheme, and weight factor adjustment, are reviewed. Finally, future trends in this field is discussed, and several promising research topics are highlighted.**

***Index Terms*— Computational burden, finite control set (FCS), model predictive control (MPC), permanent magnet synchronous motor (PMSM), robust operation, switching frequency.**

I. Introduction

ELECTRICAL machines consume about 46% of total electricity generated worldwide, resulting in about 6040 Mega-tonnes of CO_2 emission. This is by far the largest portion of electricity use. Hence, motor energy efficiency is crucial for energy conservation, environment protection, and global sustainable development. Consequentially, high-efficiency motors will dominate the market development of electrical machines worldwide [1].

Manuscript received April 27, 2022; revised June 6, 2022; accepted June 25, 2022. This work was supported in part by the National Natural Science Foundation of China (51875261), the Postgraduate Research and Practice Innovation Program of Jiangsu Province (KYCX21_3331), and the Faculty of Agricultural Equipment of Jiangsu University (NZXB20210103). Recommended by Associate Editor Derui Ding. *(Corresponding author: Xiaodong Sun.)*

Citation: T. Li, X. D. Sun, G. Lei, Z. B. Yang, Y. G. Guo, and J. G. Zhu, "Finite-control-set model predictive control of permanent magnet synchronous motor drive systems — An overview," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 12, pp. 2087–2105, Dec. 2022.

T. Li and X. D. Sun are with the Automotive Engineering Research Institute, Jiangsu University, Zhenjiang 212013, China (e-mail: ujs_liteng@163.com; xdsun@ujs.edu.cn).

G. Lei and Y. G. Guo are with the School of Electrical and Data Engineering, University of Technology Sydney, Sydney NSW 2007, Australia (e-mail: gang.lei@uts.edu.au; youguang.guo1@uts.edu.au).

Z. B. Yang is with the School of Electrical and Information Engineering, Jiangsu University, Zhenjiang 212013, China (e-mail: zbyang@ujs.edu.cn).

J. G. Zhu is with the School of Electrical and Information Engineering, University of Sydney, Sydney NSW 2006, Australia (e-mail: jianguo.zhu@sydney.edu.au).

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Digital Object Identifier 10.1109/JAS.2022.105851

Due to the advantages of small size, high power factor, high efficiency, and high reliability [2]−[5], permanent magnet synchronous motors (PMSMs) are widely used in various drive systems, such as new energy vehicle drive systems, high-speed rail traction systems, electric ship propulsion systems, and high-end CNC (computer numerical control) machine tool feed drive systems [6]−[8]. The complex working conditions in industrial productions put forward strict requirements on the control strategy of the motors.

The common control methods of PMSM drive systems are current vector control [9], [10] and direct torque control (DTC) [11]−[13]. However, the linear PI (proportional-integral) controller used in current vector control still has some defects in driving situations requiring no overshoot and high dynamic performance. The traditional DTC has the problem of large torque fluctuation at low speeds. Therefore, scholars continue to explore advanced and effective control strategies to improve the performance of PMSM drive systems, such as internal model control [14]−[17], sliding mode control [18], [19], state feedback control [20]−[23] and model predictive control (MPC) [24]−[27]. At present, power converters and transmissions are hybrid non-linear systems. The input signals of power electronic devices are discrete, which control the states (turn-on or turn-off) of the switches in each device. When designing a control scheme, there are always some constraints, such as maximum output voltage and overcurrent protection. Meanwhile, the control strategies on digital control platforms all run in discrete forms. When designing a control system, the device's model must be fully considered to adjust the parameters of the controller, provided that the model is widely known. In addition, the computing power of the control platform has been improved significantly with the rapid progress of digital microprocessors, which can fully satisfy the computation demand of MPC. The above factors, as shown in Fig. 1, have formed a strong demand for MPC among all control strategies.

The MPC algorithm originated in the late 1970s. Rodriguez took the lead in applying MPC to the PMSM drive system, setting off a global research boom [28]−[32]. MPC stands out among many advanced motor control algorithms with its multi-objective, multi-variable and multi-constraint control characteristics, as well as intuitive and simple design methods. In terms of different control variables, MPC strategies can be mainly classified into model predictive current control

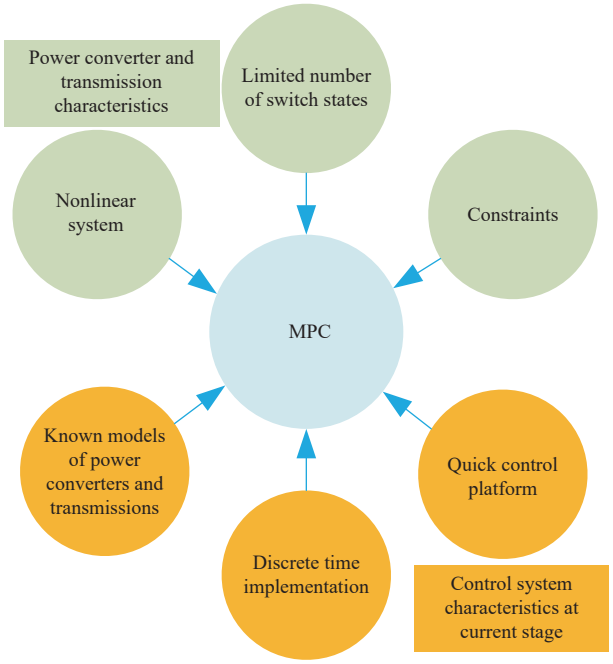


Fig. 1. MPC and its relationship with power converters, transmissions, and current control systems.

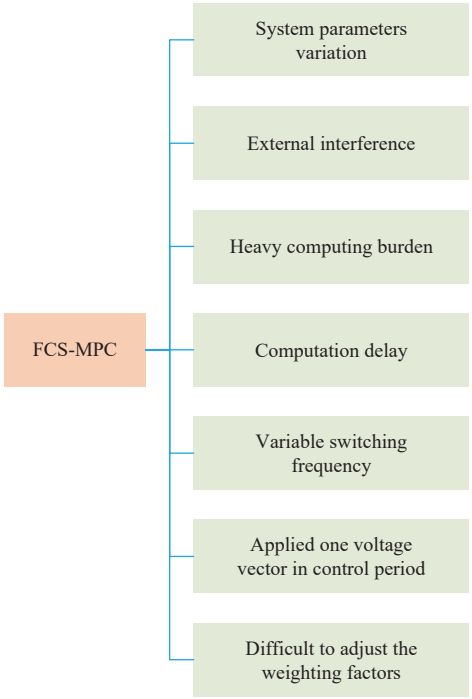


Fig. 2. Main issues in current MPC applications.

(MPCC) [2], [33], [34], model predictive torque control (MPTC) [35]−[37], model predictive speed control (MPSC) [38]−[41] and model predictive flux control (MPFC) [42], [43]. According to different control actions, the MPC strategies of PMSM can be divided into two types: finite-control-set model predictive control (FCS-MPC) [44], [45] and continuous-control-set model predictive control (CCS-MPC) [46], [47]. The main difference between them is whether the control system needs a modulation unit [48]. Among them, FCS-MPC considers the discrete switching characteristics of the converter in the prediction model and directly uses the inverter switching signal as the control action to realize the fast transient response of the system, without the need for modulation technology. Moreover, FCS-MPC has the advantages of intuitive concept, easy handling of nonlinear constraints, and realization of multivariable control [49], [50]. These remarkable advantages make it a hot spot in the MPC of power electronic systems.

However, FCS-MPC also faces many challenges in the application of PMSM drive systems, as shown in Fig. 2. For example, the non-fixation of the switching frequency [51] results in a diffused harmonic spectrum. Compared to FCS-MPC, current control techniques using modulation techniques have a fixed switching frequency, so their harmonic spectrum is concentrated at the carrier frequency and around the multiple. Meanwhile, the delays caused by online calculations [52], and model parameter mismatches will affect control performance [53]. These issues limit further development of FCS-MPC in PMSMs.

Related review studies on MPC have included the following work. In [54], the authors present the current state of MPC for power converters and drives, where relevant issues studied are cost function selection, weighting factor design, reduction in computational cost, and extension of prediction horizons. In [55], the author summarizes the basic principles of converter level and grid-level MPC. Then, a comprehensive survey is performed on the three layers of the hierarchical control architecture, including power converter control, frequency/voltage recovery, and fuel management/economic optimization. A challenge for predictive models, sampling intervals, stability and cost function design is briefly discussed. In [31], the authors focus on the following aspects: weighting factor calculation when multiple objectives are used in the cost function, current/torque harmonic distortion optimization when reducing the switching frequency of the power converter, and robustness under parameter uncertainty improvement.

The above literature provides a good summary of the development and trends of MPC. But unlike the above literature, this paper aims to provide a comprehensive and systematic review of the latest FCS-MPC strategies for PMSMs. This paper analyzes the shortcomings of traditional FCS-MPC point by point, discusses effective solutions, and summarizes and categorizes the robust operation of FCS-MPC and the reduction of computational burden. Compared with the above literature discussion, the analysis of multi-vector control strategy and delay compensation is also added. In addition, new trends in the application of FCS-MPC control schemes for PMSMs are also explored, such as the FCS-MPC scheme combined with fault-tolerant control, sensorless control, and so on.

The remainder of the paper is structured as follows. Section II summarizes the basic principle of FCS-MPC in PMSM drive systems. Section III provides an overview of recent research on the robustness of FCS-MPC in terms of three aspects. Sections IV and V provide a comprehensive review of the com-

putational burden and switching frequency of conventional FCS-MPC, respectively. Section VI investigates other FCS-MPC performance optimizations. Section VII points out promising future trends, followed by conclusions.

II. Principle of Finite-Control-Set Model Predictive Control

The conventional FCS-MPC is based on the characteristics of the limited switching states of the converter. Based on the discrete mathematical model of the converter, the converter system output under the respective effects of all the switching states is calculated through traversal. Finally, the optimal switching state is chosen to minimize the cost function and act on the system directly.

Fig. 3 shows the limited switching states produced by the two-level voltage source inverter (2L-VSI) most commonly used in PMSMs.

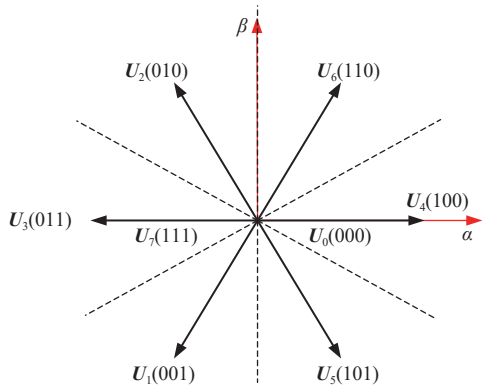


Fig. 3. Voltage vectors space of 2L-VSI.

Similar to conventional MPC, FCS-MPC also has three basic components: predictive model, objective function, and optimization process. In this section, the FCS-MPC method applied to the PMSM drive systems will be studied.

A. The Predictive Model

The continuous-time PMSM model can be written in the *d*-*q* coordinate system, and the electrical equation is

$$\begin{cases} u_d = Ri_d + L_d\dfrac{di_d}{dt} - \omega_e L_q i_q \\ u_q = Ri_q + L_q\dfrac{di_q}{dt} + \omega_e L_d i_d + \omega_e \psi_f \end{cases} \tag{1}$$

where u_d, u_q, i_d, i_q, L_d and L_q are the *d*- and *q*-axis components of the stator voltage, current, and inductance in the rotor reference frame, respectively. R_s, ψ_f and ω_e represent stator resistance, flux linkage and electrical rotor speed, respectively.

The torque equation can be expressed as

$$T_e = \frac{3}{2} n_p i_q \left[i_d \left(L_d - L_q \right) + \psi_f \right] \tag{2}$$

where n_p is the number of pole pairs of the motor.

The mechanical motion equations of the motor are written as follows:

$$J\frac{d\omega_m}{dt} = T_e - T_L - B_m\omega_m \tag{3}$$

$$\omega_e = n_p\omega_m \tag{4}$$

where J is the system inertia and B_m is the system friction coefficient. T_L and ω_m represent load torque and mechanical speed, respectively.

For common current control, (1) can be rewritten in state space form [56], [57]

$$\begin{cases} \dfrac{dx(t)}{dt} = Ax(t) + Bu(t) + C \\ y = Dx(t) \end{cases} \tag{5}$$

where the controllable state variable $x = \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}$, the control input variable $u = \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix}$, y is the controlled output variable, and

$$A = \begin{bmatrix} -\dfrac{R_s}{L_d} & \dfrac{L_q}{L_d}\omega_e \\ -\omega_e\dfrac{L_d}{L_q} & -\dfrac{R_s}{L_q} \end{bmatrix},\ B = \begin{bmatrix} \dfrac{1}{L_d} & 0 \\ 0 & \dfrac{1}{L_q} \end{bmatrix}$$

$$C = \begin{bmatrix} 0 \\ -\omega_e\dfrac{\psi_f}{L_q} \end{bmatrix},\ D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

According to the forward Euler equation, (5) can be written in discrete form as follows:

$$\begin{cases} x(k+1) = A_0x(k) + B_0u(k) + C_0 \\ y(k+1) = Dx(k+1) \end{cases} \tag{6}$$

where k and $k+1$ represent the value of the current moment and the next moment, respectively, and

$$A_0 = \begin{bmatrix} 1 - \dfrac{T_sR_s}{L_d} & T_s\omega_e\dfrac{L_q}{L_d} \\ -T_s\omega_e\dfrac{L_d}{L_q} & 1 - \dfrac{R_sT_s}{L_q} \end{bmatrix},\ B_0 = \begin{bmatrix} \dfrac{T_s}{L_d} & 0 \\ 0 & \dfrac{T_s}{L_q} \end{bmatrix}$$

$$C_0 = \begin{bmatrix} 0 \\ -\dfrac{T_s\psi_f\omega_e}{L_q} \end{bmatrix}$$

where T_s is the sampling time.

B. The Objective Function

Unless otherwise specified, the conventional FCS-MPC introduced in this paper adopts a one-step prediction.

Because the sampling time is very short, it can be considered that the reference value of the current time $x^*(k)$ is equal to the reference value of the next time $x^*(k+1)$. The objective function can be defined as follows:

$$J = \lambda_1 \left\| x^*(k) - x(k+1) \right\|_2 + \lambda_2 \|\Delta j\|_2 + \cdots \tag{7}$$

where λ_1, λ_2 are the weighting coefficients of control objectives, and Δj are the other control constraints. The multi-constraint handling nature of FCS-MPC is one of the main factors that differentiate it from many other control schemes. In

PMSM drive systems, common multi-constraint problems include consideration of voltage saturation, current limit constraints, switching frequency, and stability control [56], [58]−[61]. However, more constraints will result in several challenging issues, such as the efficient balance of various control objectives and adjusting of the weight coefficients.

C. The Optimization Process

The conventional FCS-MPC makes n predictions in one control cycle. Taking 2L-VSI as an example, there are 8 switching states S_n $(n = 1, 2, \ldots, 8)$ and these switching states to calculate the predicted value $x_n(k+1)$. Finally, the optimal switching pulse sequence S_{abc} is output to the inverter by minimizing the objective function. Similarly, in the next control loop, the above steps are repeated, and the optimization process of each loop does not correlate.

The ergodic calculations and prediction bring a great computational burden and put forward high requirements on the microprocessor. Meanwhile, the delay caused by the computational burden will also deteriorate the control performance of the system. The computational burden will be heavier for multi-phase PMSM systems and more complex inverters. Fig. 4 is a control block diagram of FCS-MPC used in PMSMs.

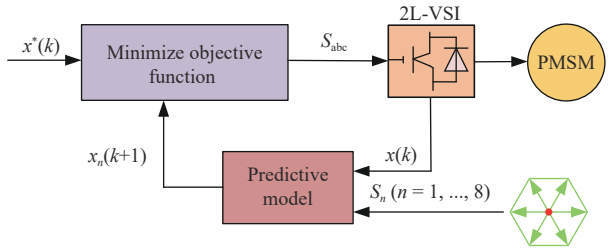


Fig. 4. A typical MPC block diagram for PMSMs.

According to the introduction of the principle of FCS-MPC, it is found that FCS-MPC has disadvantages such as dependence on parameters, a large amount of calculation, and unfixed switching frequency. With the widespread application of FCS-MPC, many scholars have made effective contributions to these issues. The following section summarizes the research status of FCS-MPC in robust operation, reducing computational burden, optimizing switching frequency, and improving control performance

III. Robust Operation of FCS-MPC

FCS-MPC is a popular and efficient control scheme. However, as shown in Fig. 5, in practical applications, PMSMs have problems including parametric uncertainties and external disturbances. Parametric uncertainties are divided into mechanical parameter changes and electrical parameter changes. Mechanical parameters refer to system inertia, while electrical parameters are resistance, inductance, and permanent magnet flux linkage. External disturbances include friction torque, load torque, and mechanical factors. The controller and sensor also have some problems, such as dead effects and measurement errors [62].

The above problems lead to inaccurate prediction models and further deteriorate control performance. Many scholars

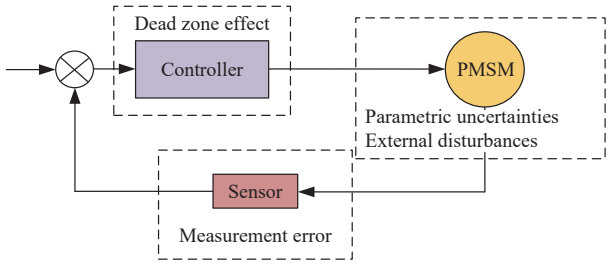


Fig. 5. Uncertainty factors of FCS-MPC scheme for PMSM.

have studied the robust operation of FCS-MPC and divided it into three aspects: disturbance estimation and compensation, prediction model improvement, and online parameter identification.

A. Disturbance Estimation and Compensation

Considering the practical applications of PMSMs, there are not only model parameter disturbances but also load disturbances and dead zone effects. Therefore, determining how to suppress the disturbance of the control systems has always been a research hotspot. Among them, observer-based disturbance estimation and compensation technology are the most widely used [63]−[66]. The basic idea is to design an observation mechanism to estimate the disturbance and uncertainty variables and then use the estimated disturbance to achieve disturbance compensation.

In [67], the proposed MPCC method took into account the perturbation of motor parameters and the demagnetization of permanent magnets compared to conventional schemes. The designed composite observer combined the sliding mode observer and Luenberger observer to observe the voltage and current values, thereby increasing the compensation voltage value to achieve robust operation. In [68], the moving horizon estimator was used to compensate for the voltage offset caused by the inductance-resistance mismatch. The influence of a permanent magnet flux linkage was eliminated in this algorithm. In [69], a second-order sliding mode observer-based on the variable gain Super-Twisting algorithm was proposed to realize parameter error estimation and one-beat delay compensation and improve the performance of FCS-MPCC in the case of five-phase permanent magnet synchronous motor model parameter mismatch. In [70], an incremental prediction model was used to exclude the effect of flux mismatch, and a sliding-mode disturbance observer was employed to estimate the predicted current value and perform delay compensation. Furthermore, a simple algorithm was developed to extract the inductance value to refresh the prediction model.

These articles mainly focus on the changes in PMSM's parameters and do not make much analysis of the inaccuracy of the predicted values caused by an external disturbance.

To suppress the unknown load torque disturbance, a load torque disturbance estimator using a two-stage extended Kalman filter (TSEKF) was proposed [71]. The proposed TSEKF estimated the system state at the same time, which greatly improved the model prediction process and enabled predictive speed control. It has a faster transient response and robustness to changes in mechanical system parameters. In

[25], in addition to using the load torque observer to estimate the actual load torque, the author also added a torque suppression factor to the cost function to avoid torque oscillation and overshoot when the speed changes rapidly. In [42], Finite-control-set model predictive torque control (FCS-MPTC) in actual operation, load disturbance and variable parameters will produce the electromagnetic torque tracking error in the torque control loop. By extending the state observer, the load disturbance was converted into a new state variable, and a disturbance compensation technique was developed through the online estimation of the lumped observer. However, the observer-based method does not eliminate the existence of model bias. It should be noted that eliminating model deviations is always the ultimate goal and reason for robust controller design.

Fig. 6 is a block diagram of the principle of compensation based on observer estimation drawn in [68]. The specific principles of various observers are not detailed here but can be referred to [57], [62]. Table I lists the disturbance/uncertainty estimation and attenuation (DUEA) techniques of AC motors transferred from [62] for readers' reference.

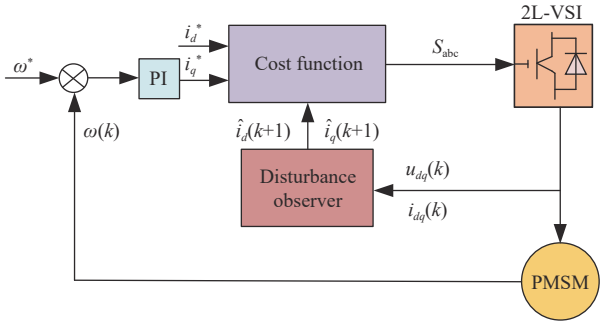


Fig. 6. Block diagram of FCS-MPC applied to PMSM based on observer estimation and compensation.

B. Improvement of the Prediction Model

The use of observers has been proved to be able to effectively resolve the impact of a single parameter mismatch. However, for PMSMs, there are multiple time-varying parameters at the same time. The impact of resistance mismatch is small, and the inductance has the greatest impact on prediction accuracy [72]. Thus, most articles only consider the influence of inductance mismatch. For this reason, many scholars generally combine improved predictive models and observers to realize the problem of multi-parameter mismatch. By improving the motor prediction model, the influence of PMSM parameter changes on the motor prediction model can be reduced. Constructing incremental predictive models and non-parametric predictive models is a common method for PMSM model predictive control model improvement, both of which improve the robustness of the system by eliminating the disturbance of motor parameters.

1) Incremental Prediction Model

In [70] and [72], an incremental prediction model was used to remove the effect of permanent magnet flux linkage. The main idea of the incremental prediction model is to expand the original model by introducing additional state variables to form an incremental expansion model. Taking the Euler discretization of the q-axis current in the d-q coordinate system of PMSM as an example, the following forward-backwards expansion is performed:

$$\begin{cases} i_q(k+1) = i_q(k) + \dfrac{T_s}{L_q}\left[-Ri_q(k) - \omega_e\left(L_d i_d(k) + \psi_f\right) \right. \\ \qquad\qquad \left. + u_q(k)\right] \\ i_q(k) = i_q(k-1) + \dfrac{T_s}{L_q}\left[-Ri_q(k-1) - \omega_e\left(L_d i_d(k-1) + \psi_f\right) \right. \\ \qquad\qquad \left. + u_q(k-1)\right]. \end{cases} \tag{8}$$

Finally, the permanent magnet flux linkage parameter is avoided in the predicted current model.

$$\begin{aligned} i_q(k+1) = &-\left[i_d(k) - i_d(k-1)\right] T_s \omega_e + \frac{T_s}{L_q}\left[u_q(k) - u_q(k-1)\right] \\ &-\left(\frac{T_s}{L_q} - 2\right) i_q(k) + \left(\frac{T_s}{L_q} - 1\right) i_q(k-1). \end{aligned} \tag{9}$$

In [73], an incremental prediction model based on MPCC was developed to eliminate permanent magnet flux linkage. Furthermore, for the disturbance of the rest of the motor parameters, a disturbance observer is used to obtain accurate inductance information. The combination of the two schemes overcomes the dependence of FCS-MPCC on the parameters of PMSMs.

However, the application of the incremental prediction model is limited to the elimination of permanent magnet flux linkage, and the other parameter mismatches still need to be combined with other robustness schemes.

2) No-Parameter Predictive Model

Nonparametric model predictive control improves the predictive robustness by eliminating parameter uncertainty and does not depend on any motor parameters.

In [74] and [75], a model-free predictive current control (MFPCC) of the PMSM drive system based on current difference detection technology was developed. The current change between two sampling moments belonging to the corresponding basic voltage vector (VV) was stored in the look-up tables (LUT). These current changes were used for current prediction and the voltage vector that causes the smallest control error was selected. The proposed MFPCC did not require information about resistance, inductance, voltage and back-EMF, so it was not affected by parameter mismatch. However, this control strategy increased the computational burden because it doubled the current sampling in one control cycle. And a higher sampling frequency was required to achieve satisfactory performance. Based on [74], a simplified MFPCC was proposed in [76], where the stator current was detected once per sampling interval instead of twice. Reference [77] addressed the stalling problem in [74], which imposed a refresh rate window at a predefined sampling period. However, applying a non-optimal switching state can adversely affect system performance. In [78], major updates were made to [74], [79]. The proposed scheme used the PMSM model to

TABLE I
COMPARISONS OF DIFFERENT DUEA METHODS REPRODUCED FROM [62]

DUEA	Complexity		Capability				
Method	Output feedbacks	Computation burden	Load torque observer	Inertial identification	Friction compensation	Torque ripple estimator	Lumped estimator
Extended state observer	Yes	Low	Yes	No	No	No	Yes
Equivalent input disturbance	Possible	Medium	Yes	No	Possible	No	Yes
Frequency-Domain disturbance observer	Yes	Low	Yes	Possible	Possible	No	Yes
Generalized proportional integral observer	Yes	Medium	Yes	No	No	No	Yes
Intelligent disturbance observer	Possible	High	Yes	Yes	Yes	Yes	Yes
Nonlinear disturbance observer	No	High	Yes	Yes	Yes	Yes	Yes
Sliding mode disturbance observer	Possible	High	Yes	Possible	Yes	Yes	Yes
Uncertainty and disturbance estimator	No	Medium	Yes	No	Possible	No	Yes
Unknown input observer	Possible	Medium	Yes	Possible	Possible	Yes	Yes

construct the equivalent differential stator currents corresponding to the seven basic voltage vectors. The current slope was defined from the current difference of two successively applied voltage vectors. The relevant update factor was introduced into the prediction to correct the mandatory response of all switch states. Regardless of whether the continuous switching state of the application was different, the scheme was applied to each current measurement to update the stored information. In [80], an improved FSC-MPCC of SPMSM was proposed. To avoid the slow current update, a novel current update mechanism was developed, which took into account the changes in resistance, rotor flux and inductance, and introduces a modified coordinate system that includes a fixed axis and a rotating axis frame. Based on this mechanism, prediction parameters can be obtained more accurately, thereby suppressing the disturbance caused by model parameter mismatch. In [81], the permanent magnet synchronous motor model mechanism was reconstructed based on the two recent current changes to overcome the stagnant current update mechanism. The proposed non-parametric predictive current control based on the current update mechanism can effectively reduce torque ripple and improve current performance.

The block diagram of the no-parameter predictive model is shown in Fig. 7.

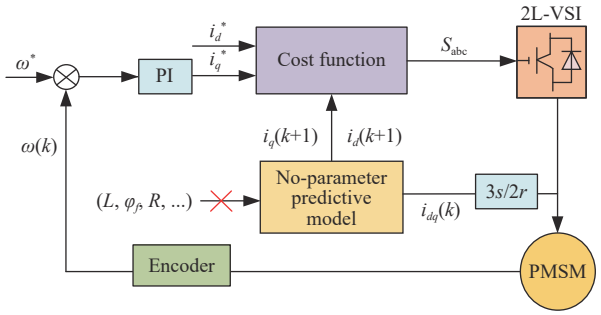


Fig. 7. Block diagram of FCS-MPC applied to PMSM based on no-parameter predictive model.

C. Online Parameter Identification

In the parameter mismatch, the influence of the inductance change is the largest, and it has a non-linear influence on the current, so the lumped compensation through the observer cannot completely restore the prediction accuracy. A more intuitive way to solve the problem is to identify the real parameters online instead of looking for steady-state equivalents [82]. Using the method of online parameter identification (OPI) to refresh the prediction model in real-time can solve the problem of the mismatch of the prediction model and realize the robust control of the system. Different from the prediction model improvement part, the online parameter identification component improves the robustness of the model by identifying the parameters online instead of eliminating the role of the parameters in the model.

The common methods of PMSM online parameter identification include the recursive least squares algorithm (RLS) [83], [84], extended Kalman filter algorithm (EKF) [53], [85], model reference adaptive algorithm (MRAS) [86], [87], and neural network algorithm (NN) [88], [89].

In [90], a discrete-time parameter identification algorithm was developed through model reference adaptive system technology, and the Popov superstability criterion was proposed to identify SPM discrete-time model parameters to realize the robust control of FCS-MPC. Reference [91] added the online least-mean-square (LMS) recognition algorithm to the control law to calculate a new system adaptive dynamic model based on the measured value. In [92], the Adaline neural network algorithm was used to identify the *d-q* axis inductance and rotor permanent magnetic flux of the IPMSM motor, combined with MPC control to achieve good steady-state and robust performance. These schemes usually use white box (WB) models with physical parameters and only identify selected subsets of these parameters online. In [93], a data-driven recursive least squares estimation method RLS with real-time capability was proposed for the current control of permanent magnet synchronous motors. The data-driven method did not use a predefined model structure composed of physical parameters and had greater flexibility than all the mentioned WB-based models. The motor model was continuously estimated online using the RLS method, and the inverter nonlinear compensation was taken into account. Table II compares several commonly used online parameter identification methods and more details can be obtained in [94],

[95]. The scheme of parameter online identification improves robustness and greatly increases the complexity and computational burden as shown in Table II.

TABLE II
COMPARISON OF OPI METHODS

Method of OPI	Speed range	Computational burden	Advantage	Disadvantage
RLS	Low to high	Medium	Easy to implement	Ill-convergence
EKF	Low to high	High	Not interfered by measurement noise	Complex calculation
MRAS	Low to high	Medium	Wide speed range application	Difficulty in gain adjustment
NN	Low to high	High	VSI nonlinearity is compensated	Long training time

The block diagram of FCS-MPC applied to PMSM based on online parameter identification is shown in Fig. 8.

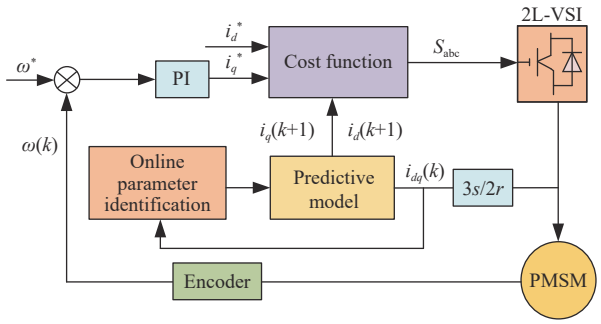


Fig. 8. Block diagram of FCS-MPC applied to PMSM based on online parameter identification.

IV. OPERATIONS TO REDUCE THE COMPUTATIONAL BURDEN

The conventional FCS-MPC needs to traverse all the basic voltage vectors, which requires a large amount of calculation and a short sampling time and limits the algorithm's use in the MPC strategy [28], [35], [96], [97]. For PMSMs, the use of more complex multi-phase or multi-level inverters will cause alternative basic voltage vectors in FCS-MPC and aggravate the calculation burden [98]–[101]. Some schemes expand the set of candidate voltage vectors to optimize steady-state performance, which will also introduce a more significant computational burden [101], [102]. The heavy computing burden has undoubtedly brought great challenges to digital chips. Therefore, more and more scholars have turned their attention to reducing complexity and reducing the burden of online computing. This paper classifies these studies into candidate vector optimization and prediction process optimization. In addition, the use of multi-step predictive control will increase computational cost, which will be described in Section VII of this paper.

A. Candidate Vector Optimization

In [103], for the six-phase PMSM control scheme, a look-up table was established to exclude useless voltage vectors in advance. This reduced the number of prediction vectors and relieved the heavy computational burden. Similarly, in [104], corresponding to the MPTC used for three-phase PMSM, the candidate set was first expanded by synthesizing virtual vectors to improve steady-state performance. Then, a look-up table was established to reduce the candidate vectors according to the relationship between the voltage vector and the torque and flux linkage. In [105], the pre-selection of the examined switching states was based on the applied state relative to the previous sampling period. This method reduces the 27 voltage vectors in the 3L-NPC to 7 and reduces the calculation amount of FCS-MPC used for PMSM. We then still need to calculate the optimal vector through the cost function with the above method.

The above candidate vector optimization scheme applies to FCS-MPC with more candidate voltage vector sets. The purpose of reducing the computational burden is to eliminate redundant or useless candidate voltage vectors through the theoretical scheme.

B. Prediction Process Optimization

The optimization of the prediction process is different from the traditional cost function calculation scheme. The optimization algorithm is used to reduce or avoid the calculation times of the cost function.

In [106], combining the deadbeat idea, a finite control set model predictive torque control (FCSMPTC) control strategy was proposed, as shown in Fig. 9. In the strategy, the accurate output voltage vector is measured and estimated first to obtain the sector where it is located and then predict the three candidate voltage vectors of the sector. U_{REF} in Fig. 10 is the estimated voltage, and the sector (DB-sector) of the voltage vector to be selected is determined according to the angle θ_{ref}. This method reduces the number of traditional optimization searches. Moreover, the duty-cycle MPC of PMSM was proposed based on the deadbeat principle [107]. After obtaining the reference voltage vector, the optimal voltage vector was determined by calculating the angle of the reference voltage vector with only one calculation. A method of using an indirect reference vector was proposed in [3]. The reference vector is determined indirectly based on the DTC theory rather than the deadbeat principle. The intensive reference vector derivation of the deadbeat principle is avoided. In [108], the fuzzy control method was used to optimize the voltage vector. The controller input was the error of the flux linkage and torque, and the output was the simplified three candidate voltage vectors. In [2] and [33], a new type of FCS-MPC applied to PMSM was proposed. The establishment of a new reference system is shown in Fig. 10, which is based on the prediction of the current trajectory when a zero voltage vector is used. The proposed method only needs to predict one zero voltage vector, not all voltage vectors. The selection of the optimal voltage vector is based on the direction of the current trajectory (i'_α, i'_β) in a given reference frame, rather than a cost function, which saves a lot of calculation time.

Table III shows the computation time in DSPACE at low speed for conventional MPCC, duty-cycle MPCC (DCMPCC) and DCMPCC based on the deadbeat principle in [107] and MPCC in [2], which is similar to that at high speed.

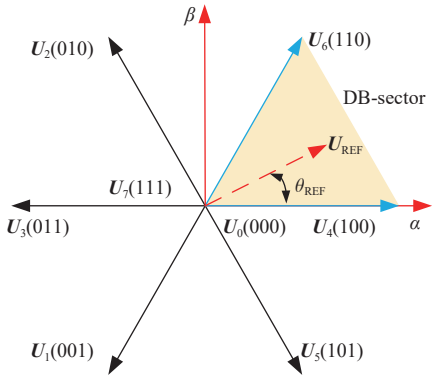


Fig. 9. 2L-VSI's deadbeat search solution.

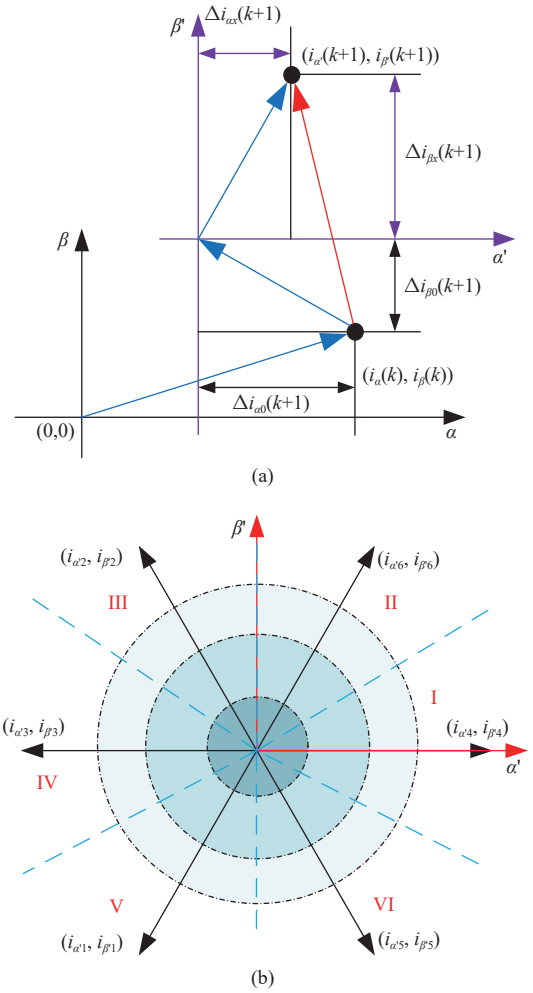


Fig. 10. The new reference frame of [24]. (a) New reference frame $\alpha'-\beta'$ based on stator reference frame $\alpha-\beta$. (b) The current loci when different active voltage vectors are applied in the reference frame $\alpha'-\beta'$.

V. Switching Frequency Optimization

Different from the classic control scheme using PWM control, FCS-MPC does not need a modulation unit because the inverter model is discretized [109], and the switching action is irregular. Hence, the switching frequency of the converter based on FCS-MPC control is not fixed, and the current or voltage harmonic spectrum are scattered [110].

The existing research scheme combines the idea of modulation to fix the switching frequency of FCS-MPC to solve this problem [111]−[113].

TABLE III
Comparison of Calculation Time

Method	MPCC	DCMPCC	DCMPCC in [107]	MPCC in [2]
Calculation time (μs)	46.2	49.3	35.4	32.1

In [114], an FCS-MPCC scheme for five-phase PMSM based on a constant switching frequency multi-vector was proposed. According to the superposition principle, the original disordered switching state was rearranged to a fixed switching frequency, and carrier-based pulse width modulation (CBPWM) was used to complete the calculation and arrangement of this complex switching sequence. Fig. 11 shows that using the method in [114], the original pulse signal is arranged into a pulse signal with a fixed switching frequency after calculating the switching time t_x ($x = a, b, c, d, e$). On the FCS-MPC control of the six-phase PMSM, a fixed switching frequency was also realized, a zero vector was inserted, and three sets of actual vectors were used to synthesize the vectors in two different ways. The switching sequence was also arranged and implemented with PWM [51]. According to the THD analysis in Fig. 12, it can be known that the THD of the conventional MPTC is 30.03%, while the proposed MPTC is 9.67%.

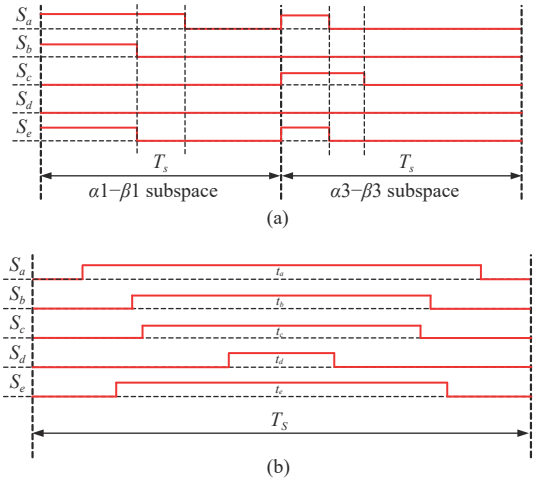


Fig. 11. Pulse generation. (a) Original switching signal; (b) Rearranged fixed switching frequency signal.

However, schemes using modulators will result in reduced transient performance.

The FCS-MPC scheme of the three-phase PMSM also achieves a fixed switching frequency. In [115], a modulation model predictive control (M2PC) scheme was proposed. The program combines SVM, MPC and FOC. While ensuring fast dynamic response, a fixed switching frequency and low torque ripple of a permanent magnet synchronous motor are obtained. Generally, M2PC is a variant of FCS-MPC. This control method aims to improve the output power quality of the system through an intrinsic modulator while retaining the advantages of FCS-MPC. The difference is that a phase is added to the modulator. M2PC predicts two active voltage vectors and

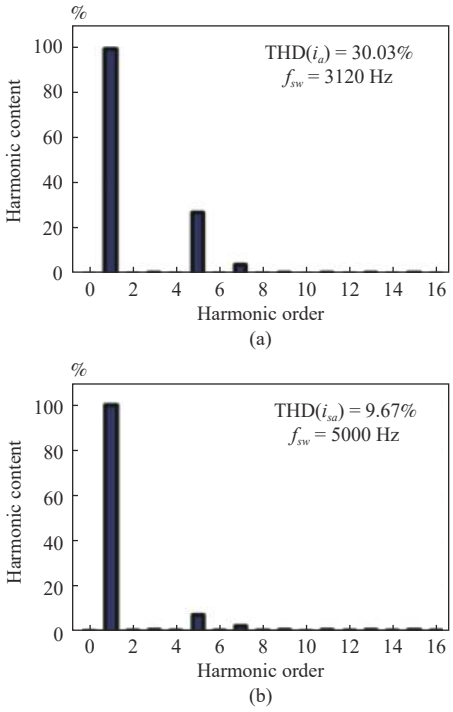


Fig. 12. Frequency spectra of the phase current i_a at 600 rpm with 4 N·m load using (a) conventional MPTC and (b) proposed MPTC [51].

determines their duty cycle instead of one voltage vector [116].

Similarly, an optimized over-modulation M2PC was discussed in [117], which can ensure the spectrum switching frequency within the linear modulation range and extend its optimized response to the over-modulation area. We use this technology to achieve a constant switching frequency, and optimized overmodulation to ensure the fastest dynamic response, to achieve the same transient performance as FCS-MPC. In [118], a low-complexity M2PC method (OST-M2PC) was proposed. Only one prediction was required per sampling period, and a sector was quickly selected by the predicted voltage to reduce computation.

Fig. 13 shows the control block diagram of a conventional M2PC. It should be noted that although M2PC is a variant of the FCS-MPC method proposed and is based on FCS-MPC while retaining the advantages of FCS-MPC. The introduction of modulators makes it closer to CCS-MPC in classification.

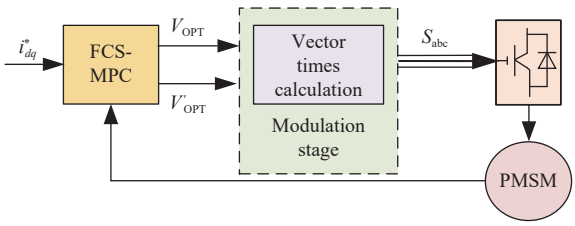


Fig. 13. The control diagram of a conventional M2PC for PMSMs [117].

VI. Other Performance Optimization Schemes

In addition to the above analysis, there are various optimization schemes to improve the performance of FCS-MPC for PMSM control. It mainly includes the following aspects, such as the application of multi-vector strategy, delay compensation, weighting factors adjustment and multi-step FCS-MPC.

A. Multi-Vector Strategy

Conventional FCS-MPC uses only one voltage vector in one control cycle, which causes large torque ripples and large current harmonics.

In [119], after selecting the best VV through the cost function, the duration was calculated by the current slope, which was responsible for reducing torque ripple. In [107], in addition to obtaining an active voltage vector from the conventional MPCC, a zero vector was also applied at the same time. A more accurate voltage selection was achieved by calculating the duration of the active voltage vector. The duty cycle was calculated by the method based on the deadbeat and current differential, respectively, which proved that the scheme based on the deadbeat was simpler and more effective.

The duty cycle control adds a zero vector in one control period to adjust the amplitude of the selected optimal active voltage vector, but it cannot adjust the phase of the selected voltage. Therefore, scholars have also studied how to adjust the phase of the applied voltage in a control cycle to improve steady-state performance.

To improve the conventional MPC with fewer active vectors, increasing the number of voltage vectors by constructing virtual vectors was proposed [104]. In total, the steady-state effect of MPTC was improved. The structure of the virtual vector was diverse and can be divided infinitely. More virtual vectors will reduce the control error and obtain better performance. There are more virtual vectors in [102] than in [104]. The researchers in [102] designed a hybrid control set (HCS) combined with virtual vectors and duty-cycle control. However, there is no doubt that both the construction of virtual vectors and the introduction of duty cycle control will make FCS-MPC more complicated, and the computational burden will be heavier.

Take the two-level VSI as an example; Fig. 14 shows the alternative voltage vector sets of the conventional FCS-MPC, the duty cycle FCS-MPC, and the FCS-MPC based on the hybrid control set. The black line represents the basic voltage vector, the blue line represents the constructed virtual vector, and the red line represents the applied voltage.

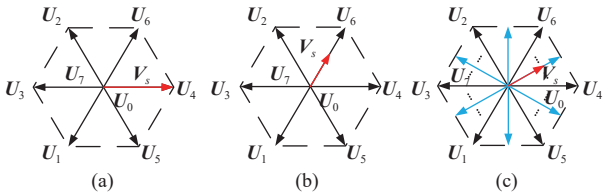


Fig. 14. Voltage vector diagram. (a) Conventional FCS-MPC; (b) Duty cycle FCS-MPC; (c) FCS-MPC based on a hybrid control set.

In addition, based on the duty cycle MPC, the selection of the second voltage vector is no longer limited to the zero vector but can also be a non-zero vector. This type of method is simpler and more effective than performing complex duty

TABLE IV
COMPARISON OF MULTI-VECTOR FCS-MPC (TAKING 2L-VSI AS AN EXAMPLE)

Item	Conventional MPC	Duty-cycle MPC	Auxiliary voltage vector MPC	Generalized double vector MPC	Three-vector MPC
Alternative voltage vectors (VVs)	8 basic VVs	6 active VVs (first choice) &2 zero VVs (second choice)	8 basic VVs + N auxiliary vectors	6 active VVs (first choice) & 8 basic VVs (second choice)	6 active VVs (first choice) & 6 active VVs (second choice) &2 zero VVs (third choice)
Voltage amplitude	Fixed	Limited Variable	Fixed	Variable	Variable
Voltage phase angle	Fixed	Fixed	Limited variable	Variable	Variable
Dynamic performance	Similar				
Steady-state performance	Improved from left to right				
Computational burden	An increase from left to right				

cycle calculations after constructing virtual vectors.

An improved dual-vector MPTC was proposed to enrich the voltage vector combination by rationally choosing the active vector instead of the zero vector in [120]. Firstly, by analyzing the torque and flux differences of a PMSM, a second vector selection table was established to determine the best VV combination to suppress torque ripple. Secondly, the duty cycle based on the torque and flux linkage differences was used to reduce the computational burden. In [121], the deadbeat scheme was used to obtain the reference voltage in advance and the second voltage vector was selected according to the sector where the reference voltage is located. The selection scheme is intuitive and simple. Calculate the duty cycle separately after obtaining the appropriate VV combination.

To further improve the steady-state performance, a three-vector model predictive control was proposed based on the two-vector model predictive control [122]−[125]. That is, three voltage vectors (two active voltage vectors and one zero vector) are used in one control cycle. By adjusting their respective duty cycles, a steady-state effect similar to SVPWM can be achieved, and the unique advantages of FCS-MPC can be maintained. Table IV compares and summarizes the above multi-vector model predictive control.

Fig. 15 is an expansion of Table IV. The verification of various schemes is performed at a sampling frequency of 10 kHz. The parameters of the permanent magnet synchronous hub motor are listed in Table V.

It is worth emphasizing that although the introduction of multi-vectors improves steady-state performance, the complexity and computational burden of FCS-MPC inevitably increases. Therefore, the research in Section IV is necessary. Fig. 16 shows the voltage vector space of the auxiliary voltage vector MPC (N = 6).

B. Delay Compensation

Various delays, including filtering, sampling and other factors, make the output of the controller have application delays. The update mechanism of modern digital signal processors and microcontrollers allows the command voltage at the current moment to be applied at the next moment. Therefore, the application of voltage is inaccurate and leads to deterioration of control performance [126]−[129]. Therefore, it is necessary to compensate for the delay in FCS-MPC to improve the control accuracy.

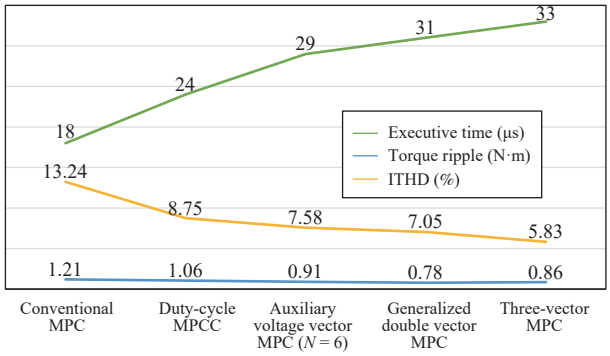


Fig. 15. Execution time, current harmonics and torque ripple under various control schemes.

TABLE V
PMSM SYSTEM PARAMETERS

Parameter	Symbol	Value
Permansent-magnet flux linkage	ψ_f	0.215 Wb
Number of pole pairs	P_n	22
Stator inductance	L_s	4.5 mH
Stator resistance	Rs	0.8 Ω
DC-link voltage	U_{dc}	420 V
Rated speed	N	360 rpm
Rated power	P_N	3 kW
Frictional coefficient	B	0.0006 N·m/s
Inertia	J	0.03 kgm^2

Fig. 17 shows the change of i_α without delay and with delay time in the static coordinate system of model predictive current control. The superscript p represents the predicted value and the subscript * represents the reference value. When there is no delay, the predicted value is consistent with the actual value. But when there is a delay, the actual value will deviate from the predicted value.

In [72], a two-step prediction was used to compensate for the delay of FCS-MPCC. Firstly, the i_s^p $(k+1)$ obtained by one-step prediction was replaced by the $i_s(k)$ in the original prediction model, and then the $i_s^p(k+2)$ was brought into the cost function as a new predicted current value. In [130], two-

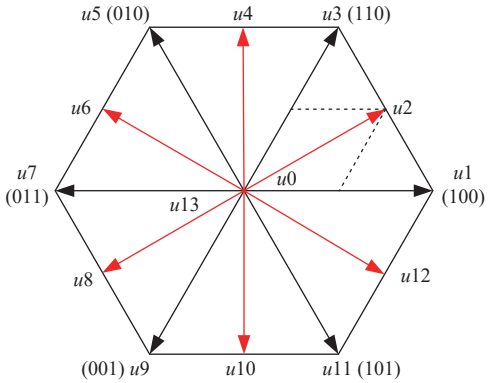


Fig. 16. Auxiliary voltage vector MPC ($N = 6$).

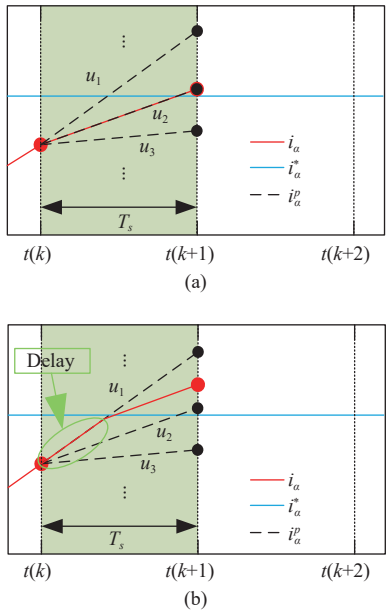


Fig. 17. Model predictive current control. (a) Without delay; (b) With delay.

step prediction was used to obtain the predicted torque and flux value of $(k+2)$-th time in FCS-MPTC. Unlike the general two-step prediction, in [107], first, the traditional one-step prediction current was compensated for using Heun's method as a new one-step prediction current. Then we changed the one-step prediction to two-step prediction to achieve delay compensation. This method has higher accuracy.

However, two-step prediction cannot accurately compensate for the deviation value. Therefore, some schemes have been studied to analyze the current value of the error for prediction and compensation. In [52], the calculation delay in FCS-MPC was described in detail. Based on the analysis of the delay, the delay time was calculated, and the corresponding current change was directly compensated to the predicted current. Compared with the traditional two-step forecasting strategy, it is simpler and we can choose the appropriate switch to optimize control performance. This method provides high-performance delay compensation technology for PMSM's MPCC. In [131], a novel accurate PMSM model based on the numerical solution was proposed to replace and eliminate the prediction error caused by Euler's dispersion. Correspondingly, delay estimation and the current compensation scheme have been studied to achieve excellent performance compensation.

C. Weighting Factor Adjustment

One of the advantages of FCS-MPC is that it can control multiple variables simultaneously with a single control rate through weighting factors. In the implementation, efficient determination of the weighting factors is crucial to the optimization of control performance [132], [133].

According to the nature of different items involved in the cost function, the cost function was divided into different groups in [134] to determine the adjustment scheme of weighting factors. Table VI is built on this. Many studies adjust the weight factor through various methods, such as the empirical method, genetic algorithm, and fuzzy decision-making, which have achieved good control performance [29], [134]–[136].

TABLE VI
COST FUNCTION CLASSIFICATION

Classification	Application	Cost function (CF)
CF without weighting factors	Current control	$J = \lvert i^* - i^p \rvert$
	Voltage control	$J = \lvert U^* - U^p \rvert$
CF with secondary items	Switching frequency reduction	$J = \lvert i^* - i^p \rvert + \lambda_{sw} f^2$
	Common mode voltage reduction	$J = \lvert i^* - i^p \rvert + \lambda_v V^2$
	Reactive power reduction	$J = \lvert i^* - i^p \rvert + \lambda_p P^2$
CF with equally important items	Torque and flux control	$J = \frac{1}{T_{en}^2}\left(T_e^* - T_e^p\right)^2 + \frac{\lambda_\psi}{\psi_{sn}^2}\left(\psi_s^* - \psi_s^p\right)^2$

Although the above methods of adjusting the weighting factor have been proven to improve the performance of the control system, their complex design affects its further use in the actual system.

Different from the traditional principle that torque and flux are equally important to adjust the weighting factor, a new cost function is proposed in [121], [137], [138], which only includes the reference voltage vector and candidate voltage vector, eliminating the step of adjusting the weighting factor. In [139], the reference torque was changed to flux linkage by using the maximum torque per ampere (MPTA) method, as shown in Fig. 18. Therefore, the control variable changed from torque and flux to single flux. This scheme eliminates the adjustment of the weighting factor but also increases the ripple of torque. Similarly, when $i_d = 0$ was used in [140], the relationship between torque and flux was transformed into flux to simplify the control variable of the cost function. In [141], a simplified cost function was proposed for multi-phase PMSM. The original cost function included torque error, flux error in the first and third space, and current limiting term. Firstly, the virtual vector was used to eliminate the harmonics in the third space to reduce the flux error term, and then the

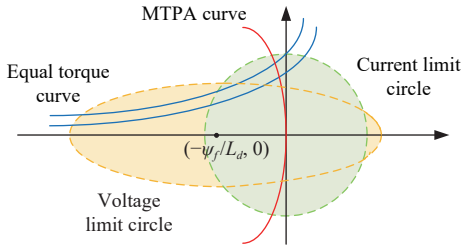


Fig. 18. The trajectory of maximum torque per ampere (MTPA) in [139].

deadbeat scheme was used to eliminate the torque error term. Finally, the addition of the weighting factor was avoided.

However, the scheme of eliminating the weighting factor is not suitable for the cost function with secondary items as shown in Table VI. Because the secondary items are not specifically related to the primary control variable.

VII. Future Trends

The FCS-MPC strategy has been successfully applied in the PMSM system in recent years. However, the actual needs of industrial applications and the pursuit of higher performance make the FCS-MPC need to be improved constantly. Here are the trends for future research.

A. Multi-Step Control Strategy

In the field of industrial production and electric vehicle, the reliability problem of power converters are becoming more and more serious due to the continuous improvement of power density. Temperature increases have the most significant effect on reliability. The most effective way to reduce the temperature rise caused by switching loss is to optimize the control mode and buffer technology. Multi-step FCS-MPC is a good choice. Most of the current research on FCS-MPC is single-step MPC. In comparison, multi-step FCS-MPC has a long-term global optimal solution rather than a local optimum of single-step FCS-MPC. Multi-step MPC was proposed many years ago [142], [143], and it has only recently gained much attention as a key driving force for improving efficiency, especially in high-power electrical drives [31]. Multi-step FCS-MPC can effectively reduce the switching frequency of inverters, greatly improving the efficiency of the motor drive system [144], [145].

Fig. 19 shows that the multi-step MPC can make the switching loss smaller under the same THD, thereby improving the efficiency of the drive system. In Fig.19, f_s is the sampling frequency and λ_u is the weighting factor [56].

Nevertheless, multi-step FCS-MPC has not been widely used because of its high computational complexity, and it is difficult to implement it in a short sampling period [146], [147]. Fig. 20 shows the search tree of a multi-step model predictive control with a predicted field of view of three as an example.

To reduce the computational complexity and improve the accuracy, researchers have made many attempts in the direction of fast prediction and intelligent prediction based on MPC. This is the future research direction of multi-step MPC.

In [148], a frequency band-based multi-step MPTC was proposed for PMSM drivers. First, the multi-step prediction

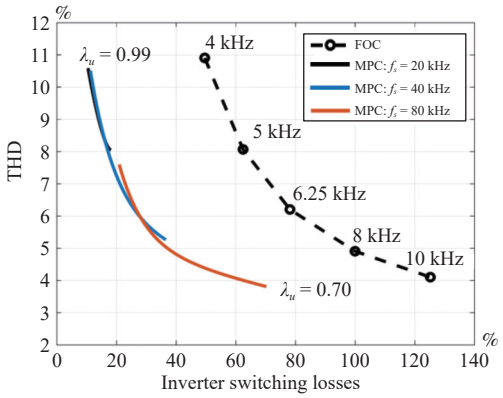


Fig. 19. The relationship between inverter switching loss and THD [56].

model was simplified, and the band constraint was used to reduce the computational complexity. Then the steady-state and transient control modes were designed to ensure good steady-state and dynamic performance. In [149], a fast-response multi-step MPTC was proposed for PMSM, and the torque was converted into the flux linkage to eliminate the weighting factor. In the transient process, Pontryagin's maximum principle was designed to obtain the minimum time flux trajectory. Along the trajectory, a one-step MPTC was used to obtain the best voltage vector. Aimed at the underlying least-squares integer programming problem, a non-recursive spherical decoding algorithm is developed to solve the multi-step MPC complex optimization problem [150]. Given the more powerful FPGA platforms that exist today, it is expected that fields of view up to a step size $N = 15$ can be achieved in real-time within a short sampling interval of 25 μs. And it is pointed out that for more complex systems, the practical implementation of multi-step FCS-MPC remains a promising topic for future research.

Furthermore, for higher power motor drives, an optimized pulse mode as shown in Section V can be combined with multi-step MPC to further reduce load current distortion while maintaining low switching frequency operation. Moreover, with the development of new digital platforms, the digital computing power is getting stronger and stronger, and multi-step FCS-MPC is bound to become the future research trend.

B. Advanced Robust FCS-MPC

The PMSM drive system will suffer from various external influences in the actual operation, and the external disturbance will produce delay, measurement error and parameter mismatch, which will worsen system performance and increase the difficulty of parameter tuning. At present, the high requirement of motor control precision makes robust model predictive control a hot topic.

Section III introduces the improvement of the prediction model, disturbance estimation compensation and parameter online identification. However, more studies are conducted to passively reduce the influence of parameter mismatch based on the construction of disturbance observer and model parameter estimation, and fewer studies are conducted on anti- disturbance performance based on the motor model itself [151], [152]. For example, in [153], the estimation scheme based on

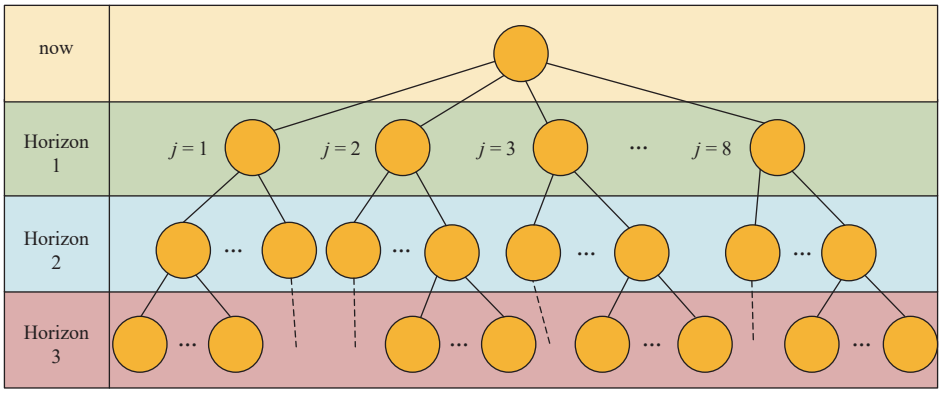


Fig. 20. Visualization of the search tree with prediction range $N = 3$.

a data-driven model was adopted to enhance the robustness of the driver to parameter uncertainty. Rather than using motor parameters directly, the inputs and outputs of the control system were used to estimate the prediction. In [154], a self-regulating strategy for the stator current mathematical model was proposed, which used the prediction error of stator current to calculate the parameter change, design the parameter change compensation strategy, and correct the mathematical model in real-time in each control cycle.

Secondly, these three schemes can enhance anti-disturbance performance. The combination of these schemes can further enhance the performance of robustness [72], [73] and improve the control accuracy of the motor.

C. Stability Analysis

Research on cost function mainly focuses on multi-constraint variables and weighting coefficient configuration, as shown in Section VI-C. However, there are few stability analyses on the control system, because most of the constructed cost functions do not conform to the definition of the Lyapunov function. Therefore, the traditional control theory cannot analyze its stability, transient response characteristics, and steady-state response characteristics. It is often verified by simulation and experiment. Therefore, the analysis theory is relatively backward and leads to limitations of the proposed solutions. Therefore, some cost functions based on the Lyapunov function stability theory are designed to guarantee the stability of the system [155], [156].

In [157], the control law based on continuous input is developed from the control-Lyapunov function of feedback control law and online adaptive law. Then the continuous control law is transformed into the relevant constraints of FCS-MPC. In the FCS-MPC of SPMSM, the stability condition based on the adaptive estimation law is maintained and further optimized by exhaustive search. In [158], an improved Lyapunov function constraint was proposed, which can allow for non-monotonic convergence to the terminal set, thus reducing the conservativeness caused by the traditional Lyapunov function constraint, thus obtaining better performance and ensuring the stability and recursive feasibility of the proposed prediction scheme. An MPTC for PMSM was designed in [159], and its cost function was designed based on factors including torque error, maximum torque per amp, and stator current constraints. The Lyapunov control theory is introduced to ensure stability when determining the weight coefficient of the cost function, to obtain the optimal voltage vector of the inverter. Compared with the conventional FCS-MPC, the torque ripple is reduced and the system robustness is significantly improved.

However, the stability of FCS-MPC in PMSM drive systems remains to be studied and needs further attention to be implemented effectively in industrial applications.

D. Combine With Other Control Schemes

It is promising to combine FCS-MPC with the actual applications of PMSM drive systems. Many control schemes, such as sensorless control and fault-tolerant control, are increasingly used with FCS-MPC to obtain better control performance.

The robustness, computational burden, steady-state and dynamic performance of the new control scheme generated by the combination of these schemes still need to be further studied. For example, both sensorless control and FCS-MPC are parameter dependent schemes, and sensorless control will introduce parameters to adjust, which will increase the computational burden. For example, Fig. 21 shows the MPCC fault-tolerant solution using virtual vectors [160]. The combination of the two saves the complexity of the conventional fault-tolerant solution and achieves good results. Fig. 22 shows the tracking error of the FCS-MPC based on the finite position set-phase locked loop (FPS-PLL).

Table VII lists some latest combined control schemes for PMSM drive systems for different applications [161]–[168].

Furthermore, complex FCS-MPC schemes may require a large amount of computational time. To avoid unnecessary resource consumption while ensuring control performance, a popular method is to combine event-triggered control and FCS-MPC [169]–[171]. Event-triggered control significantly reduces computational cost, making it possible to develop more sophisticated solutions.

VIII. Conclusion

This paper reviewed the state-of-art FCS-MPC method for PMSM drive systems. The basic principles and three elements of FCS-MPC were reviewed. Some problems to be solved and related research results of FCS-MPC were summarized, such as robust operation, reduction of the computa-

TABLE VII
SYNTHESIS OF FCS-MPC AND OTHER CONTROL SCHEMES FOR PMSM DRIVE SYSTEMS

Problem description	Control scheme	Related references
Open circuit fault tolerance	FCS-MPC+Fault tolerant control	[162]–[164]
High cost and high failure rate of mechanical sensors	Sensorless control+FCS-MPC	[161], [165], [166]
Velocity tracking problem	FCS-MPC+Active disturbance rejection control	[167], [168]
Heavy computational burden	FCS-MPC+Event triggered control	[169]

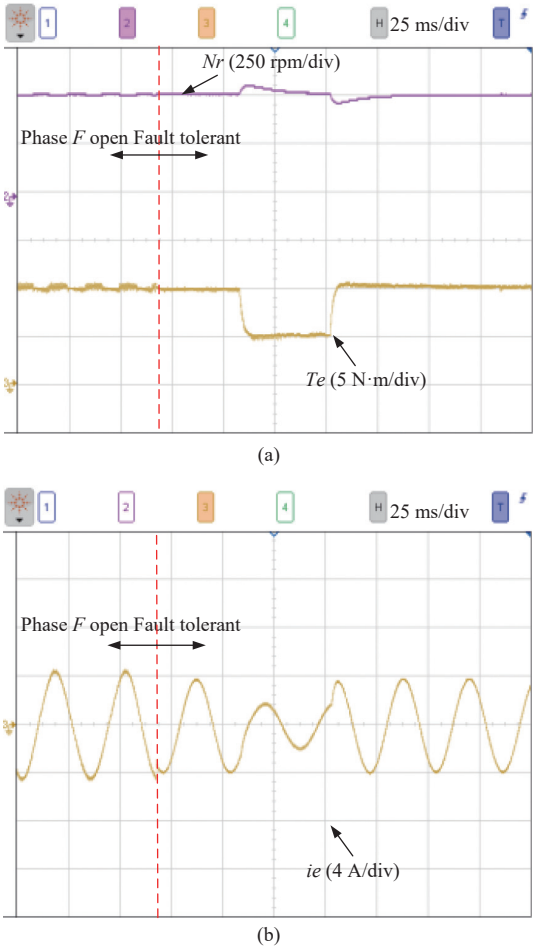


Fig. 21. Load torque variation test based on MPCC fault-tolerant control. (a) Torque and speed; and (b) E-phase current. [160].

tional burden, switching frequency optimization, and other optimization schemes like multi-vector strategies, delay compensation and weighting factor adjustment. Several new research ideas and promising development directions were proposed. To the best knowledge of the authors, the above-discussed issues of FCS-MPC still need further development and research, and the analysis theory applicable to FCS-MPC should be further improved. Multi-step FCS-MPC will gain more and more attention. The vigorous development of new methods makes it possible for the PMSM drive systems to combine solutions such as fault-tolerant control and sensorless control in practical applications. Further research is needed to explore the combination of these methods for better results.

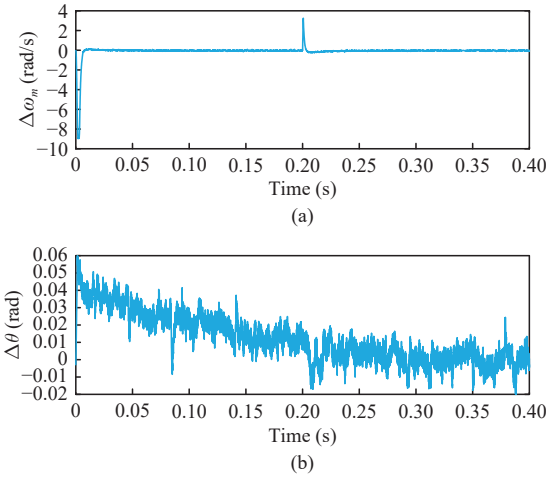


Fig. 22. The FCS-MPC with FPS-PLL. (a) Angular velocity tracking error; and (b) Angle tracking error [161].

REFERENCES

[1] G. Lei, J. G. Zhu, Y. G. Guo, C. C. Liu, and B. Ma, "A review of design optimization methods for electrical machines," *Energies*, vol. 10, no. 12, p. 1962, Nov. 2017.

[2] X. D. Sun, M. K. Wu, G. Lei, Y. G. Guo, and J. G. Zhu, "An improved model predictive current control for PMSM drives based on current track circle," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 3782–3793, May 2021.

[3] S. X. Niu, Y. X. Luo, W. N. Fu, and X. D. Zhang, "An indirect reference vector-based model predictive control for a three-phase PMSM motor," *IEEE Access*, vol. 8, pp. 29435–29445, Jan. 2020.

[4] Y. H. Hwang and J. Lee, "HEV motor comparison of IPMSM with Nd sintered magnet and heavy rare-earth free injection magnet in the same size," *IEEE Trans. Appl. Supercond.*, vol. 28, no. 3, p. 5206405, Apr. 2018.

[5] X. D. Sun, Z. J. Jin, Y. F. Cai, Z. B. Yang, and L. Chen, "Grey wolf optimization algorithm based state feedback control for a bearingless permanent magnet synchronous machine," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 13631–13640, Dec. 2020.

[6] X. F. Ding, H. Guo, R. Xiong, F. D. Chen, D. H. Zhang, and C. Gerada, "A new strategy of efficiency enhancement for traction systems in electric vehicles," *Appl. Energy*, vol. 205, pp. 880–891, Nov. 2017.

[7] E. Trancho, E. Ibarra, A. Arias, I. Kortabarria, P. Prieto, I. Martínez De Alegría, J. Andreu, and I. López, "Sensorless control strategy for light-duty EVs and efficiency loss evaluation of high frequency injection under standardized urban driving cycles," *Appl. Energy*, vol. 224, pp. 647–658, Aug. 2018.

[8] X. D. Sun, Z. Shi, Y. F. Cai, G. Lei, Y. G. Guo, and J. G. Zhu, "Driving-cycle-oriented design optimization of a permanent magnet hub motor drive system for a four-wheel-drive electric vehicle," *IEEE Trans. Transp. Electrific.*, vol. 6, no. 3, pp. 1115–1125, Sept. 2020.

[9] H. Yao, Y. Yan, T. N. Shi, G. Z. Zhang, Z. Q. Wang, and C. L. Xia, "A novel svpwm scheme for field-oriented vector-controlled PMSM drive system fed by cascaded H-bridge inverter," *IEEE Trans. Power Electron.*, vol. 36, no. 8, pp. 8988–9000, Aug. 2021.

[10] Q. F. Zhang, H. H. Guo, C. Guo, Y. C. Liu, D. Wang, K. Y. Lu, Z. R. Zhang, X. Z. Zhuang, and D. Z. Chen, "An adaptive proportional-integral-resonant controller for speed ripple suppression of PMSM drive due to current measurement error," *Int. J. Electr. Power Energy Syst.*, vol. 129, p. 106866, Jul. 2021.

[11] S. Vaez-Zadeh, "Analysis of a DTC with back emf oriented voltage for PMS motor drives," *IEEE Trans. Energy Convers.*, vol. 33, no. 3, pp. 1594–1596, Sept. 2018.

[12] Y. Z. Zhou, X. G. Lin, and M. Cheng, "A fault-tolerant direct torque control for six-phase permanent magnet synchronous motor with arbitrary two opened phases based on modified variables," *IEEE Trans. Energy Convers.*, vol. 31, no. 2, pp. 549–556, Jun. 2016.

[13] X. D. Sun, K. K. Diao, G. Lei, Y. G. Guo, and J. G. Zhu, "Direct torque control based on a fast modeling method for a segmented-rotor switched reluctance motor in HEV application," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 1, pp. 232–241, Feb. 2021.

[14] X. D. Sun, L. Chen, H. B. Jiang, Z. B. Yang, J. F. Chen, and W. Y. Zhang, "High-performance control for a bearingless permanent-magnet synchronous motor using neural network inverse scheme plus internal model controllers," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3479–3488, Jun. 2016.

[15] Z. W. Ping, Q. C. Ma, T. Wang, Y. Z. Huang, and J. G. Lu, "Speed tracking control of permanent magnet synchronous motor by a novel two-step internal model control approach," *Int. J. Control Autom. Syst.*, vol. 16, no. 6, pp. 2754–2762, Dec. 2018.

[16] X. D. Sun, Z. Shi, L. Chen, and Z. B. Yang, "Internal model control for a bearingless permanent magnet synchronous motor based on inverse system method," *IEEE Trans. Energy Convers.*, vol. 31, no. 4, pp. 1539–1548, Dec. 2016.

[17] Z. W. Ping, T. Wang, Y. Z. Huang, H. Wang, J. G. Lu, and Y. Y. Li, "Internal model control of PMSM position servo system: Theory and experimental results," *IEEE Trans. Ind. Inf.*, vol. 16, no. 4, pp. 2202–2211, Apr. 2020.

[18] P. Gao, G. M. Zhang, H. M. Ouyang, and L. Mei, "An adaptive super twisting nonlinear fractional order PID sliding mode control of permanent magnet synchronous motor speed regulation system based on extended state observer," *IEEE Access*, vol. 8, pp. 53498–53510, Mar. 2020.

[19] Z. J. Jin, X. D. Sun, G. Lei, Y. G. Guo, and J. G. Zhu, "Sliding mode direct torque control of SPMSMS based on a hybrid wolf optimization algorithm," *IEEE Trans. Ind. Electron.*, vol. 69, no. 5, pp. 4534–4544, May 2022.

[20] X. D. Sun, M. K. Wu, Z. B. Yang, G. Lei, and Y. G. Guo, "High-performance control for a permanent-magnet linear synchronous generator using state feedback control scheme plus grey wolf optimisation," *IET Electr. Power Appl.*, vol. 14, no. 5, pp. 771–780, May 2020.

[21] T. Tarczewski and L. M. Grzesiak, "Constrained state feedback speed control of PMSM based on model predictive approach," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3867–3875, Jun. 2016.

[22] Z. G. Yin, Y. X. Gu, C. Du, and F. T. Gao, "Research on back-stepping control of permanent magnet linear synchronous motor based on extended state observer," in *Proc. IEEE Int. Power Electronics and Application Conf. and Expo.*, Shenzhen, China, 2018, pp. 179–183.

[23] X. D. Sun, C. C. Hu, G. Lei, Y. G. Guo, and J. G. Zhu, "State feedback control for a PM hub motor based on gray wolf optimization algorithm," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 1136–1146, Jan. 2020.

[24] X. F. Wang, X. C. Fang, S. Lin, F. Lin, and Z. Yang, "Predictive common-mode voltage suppression method based on current ripple for permanent magnet synchronous motors," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 7, no. 2, pp. 946–955, Jun. 2019.

[25] M. Liu, K. W. Chan, J. F. Hu, W. Z. Xu, and J. Rodriguez, "Model predictive direct speed control with torque oscillation reduction for PMSM drives," *IEEE Trans. Ind. Inf.*, vol. 15, no. 9, pp. 4944–4956, Sept. 2019.

[26] S. Chai, L. Wang, and E. Rogers, "A cascade MPC control structure for a PMSM with speed ripple minimization," *IEEE Trans. Ind. Electron.*, vol. 60, no. 8, pp. 2978–2987, Aug. 2013.

[27] X. D. Sun, C. C. Hu, J. G. Zhu, S. H. Wang, W. Q. Zhou, Z. B. Yang, G. Lei, K. Li, B. Zhu, and Y. G. Guo, "MPTC for PMSMs of EVs with multi-motor driven system considering optimal energy allocation," *IEEE Trans. Magn.*, vol. 55, no. 7, p. 8104306, Jul. 2019.

[28] J. Rodriguez, M. Kazmierkowski, J. R. Espinoza, Zanchetta, H. Abu-Rub, H. A. Young, and C. A. Rojas, "State of the art of finite control set model predictive control in power electronics," *IEEE Trans. Ind. Inf.*, vol. 9, no. 2, pp. 1003–1016, May 2013.

[29] F. Villarroel, J. R. Espinoza, C. A. Rojas, J. Rodriguez, M. Rivera, and D. Sbarbaro, "Multiobjective switching state selector for finite-states model predictive control based on fuzzy decision making in a matrix converter," *IEEE Trans. Ind. Electron.*, vol. 60, no. 2, pp. 589–599, Feb. 2013.

[30] J. L. Elizondo, A. Olloqui, M. Rivera, M. E. Macias, O. Probst, O. M. Micheloud, and J. Rodriguez, "Model-based predictive rotor current control for grid synchronization of a DFIG driven by an indirect matrix converter," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 2, no. 4, pp. 715–726, Dec. 2014.

[31] J. Rodriguez, C. Garcia, A. Mora, F. Flores-Bahamonde, Acuna, M. Novak, Y. C. Zhang, L. Tarisciotti, S. A. Davari, Z. B. Zhang, F. X. Wang, M. Norambuena, T. Dragicevic, F. Blaabjerg, T. Geyer, R. Kennel, D. A. Khaburi, M. Abdelrahem, Z. Zhang, N. Mijatovic, and R. Aguilera, "Latest advances of model predictive control in electrical drives — Part I: Basic concepts and advanced strategies," *IEEE Trans. Power Electron.*, vol. 37, no. 4, pp. 3927–3942, Apr. 2022.

[32] J. Rodriguez, C. Garcia, A. Mora, S. A. Davari, J. Rodas, D. F. Valencia, M. Elmorshedy, F. X. Wang, K. K. Zuo, L. Tarisciotti, F. Flores-Bahamonde, W. Xu, Z. B. Zhang, Y. C. Zhang, M. Norambuena, A. Emadi, T. Geyer, R. Kennel, T. Dragicevic, D. A. Khaburi, Z. Zhang, M. Abdelrahem, and N. Mijatovic, "Latest advances of model predictive control in electrical drives. Part II: Applications and benchmarking with classical control methods," *IEEE Trans. Power Electron.*, vol. 37, no. 5, pp. 5047–5061, May 2022.

[33] D. D. Su, C. N. Zhang, and Y. G. Dong, "Finite-state model predictive current control for surface-mounted permanent magnet synchronous motors based on current locus," *IEEE Access*, vol. 5, pp. 27366–27375, Nov. 2017.

[34] F. X. Wang, S. H. Li, X. Z. Mei, W. Xie, J. Rodríguez, and R. M. Kennel, "Model-based predictive direct control strategies for electrical drives: An experimental evaluation of PTC and PCC methods," *IEEE Trans. Ind. Inf.*, vol. 11, no. 3, pp. 671–681, Jun. 2015.

[35] M. Preindl and S. Bolognani, "Model predictive direct torque control with finite control set for PMSM drive systems, Part 1: Maximum torque per ampere operation," *IEEE Trans. Ind. Inf.*, vol. 9, no. 4, pp. 1912–1921, Nov. 2013.

[36] O. Sandre-Hernandez, J. De Jesus Rangel-Magdaleno, and R. Morales-Caporal, "Modified model predictive torque control for a PMSM-drive with torque ripple minimisation," *IET Power Electron.*, vol. 12, no. 5, pp. 1033–1042, May 2019.

[37] X. D. Sun, M. K. Wu, C. F. Yin, and S. H. Wang, "Model predictive thrust force control for linear motor actuator used in active suspension," *IEEE Trans. Energy Convers.*, vol. 36, no. 4, pp. 3063–3072, Dec. 2021.

[38] Q. Fei, Y. T. Deng, H. W. Li, J. Liu, and M. Shao, "Speed ripple minimization of permanent magnet synchronous motor based on model predictive and iterative learning controls," *IEEE Access*, vol. 7, pp. 31791–31800, Mar. 2019.

[39] E. J. Fuentes, C. A. Silva, and J. I. Yuz, "Predictive speed control of a two-mass system driven by a permanent magnet synchronous motor," *IEEE Trans. Ind. Electron.*, vol. 59, no. 7, pp. 2840–2848, Jul. 2012.

[40] Kakosimos and H. Abu-Rub, "Predictive speed control with short prediction horizon for permanent magnet synchronous motor drives," *IEEE Trans. Power Electron.*, vol. 33, no. 3, pp. 2740–2750, Mar. 2018.

[41] M. Preindl and S. Bolognani, "Model predictive direct speed control with finite control set of PMSM drive systems," *IEEE Trans. Power Electron.*, vol. 28, no. 2, pp. 1007–1015, Feb. 2013.

[42] L. M. Yan, M. F. Dou, and Z. G. Hua, "Disturbance compensation-based model predictive flux control of SPMSM with optimal duty cycle,"*IEEE J. Emerg. Sel. Top. Power Electron.*,vol. 7,no. 3,pp. 1872–1882, Sept. 2019.

[43] S. D. Huang, G. Wu, F. Rong, C. F. Zhang, S. Huang, and Q. W. Wu, "Novel predictive stator flux control techniques for PMSM drives," *IEEE Trans. Power Electron.*, vol. 34, no. 9, pp. 8916–8929, Sept. 2019.

[44] X. D. Sun, T. Li, M. Yao, G. Lei, Y. G. Guo, and J. G. Zhu, "Improved

finite-control-set model predictive control with virtual vectors for PMSHM drives," *IEEE Trans. Energy Convers.*, vol. 37, no. 3, pp. 1885–1894, Sept. 2022.

[45] F. Yu, X. Liu, Z. H. Zhu, and J. F. Mao, "An improved finite-control-set model predictive flux control for asymmetrical six-phase PMSMS with a novel duty-cycle regulation strategy," *IEEE Trans. Energy Convers.*, vol. 36, no. 2, pp. 1289–1299, Jun. 2021.

[46] Y. N. Zhou, H. M. Li, R. D. Liu, and J. K. Mao, "Continuous voltage vector model-free predictive current control of surface mounted permanent magnet synchronous motor," *IEEE Trans. Energy Convers.*, vol. 34, no. 2, pp. 899–908, Jun. 2019.

[47] F. X. Wang, L. He, and J. Rodriguez, "FPGA-based continuous control set model predictive current control for PMSM system using multistep error tracking technique," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 13455–13464, Dec. 2020.

[48] A. A. Ahmed, B. K. Koh, and Y. I. Lee, "A comparison of finite control set and continuous control set model predictive control schemes for speed control of induction motors," *IEEE Trans. Ind. Inform.*, vol. 14, no. 4, pp. 1334–1346, Apr. 2018.

[49] S. Kouro, M. A. Perez, J. Rodriguez, A. M. Llor, and H. A. Young, "Model predictive control: MPC's role in the evolution of power electronics," *IEEE Ind. Electron. Mag.*, vol. 9, no. 4, pp. 8–21, Dec. 2015.

[50] C. S. Lim, E. Levi, M. Jones, N. A. Rahim, and W. Hew, "FCS-MPC-based current control of a five-phase induction motor and its comparison with PI-PWM control," *IEEE Trans. Ind. Electron.*, vol. 61, no. 1, pp. 149–163, Jan. 2014.

[51] Y. X. Luo and C. H. Liu, "Multi-vector-based model predictive torque control for a six-phase PMSM motor with fixed switching frequency," *IEEE Trans. Energy Convers.*, vol. 34, no. 3, pp. 1369–1379, Sept. 2019.

[52] J. Q. Gao, C. Gong, W. Z. Li, and J. L. Liu, "Novel compensation strategy for calculation delay of finite control set model predictive current control in PMSM," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5816–5819, Jul. 2020.

[53] S. Ichikawa, M. Tomita, S. Doki, and S. Okuma, "Sensorless control of permanent-magnet synchronous motors using online parameter identification based on system identification theory," *IEEE Trans. Ind. Electron.*, vol. 53, no. 2, pp. 363–372, Apr. 2006.

[54] S. Vazquez, J. Rodriguez, M. Rivera, L. G. Franquelo, and M. Norambuena, "Model predictive control for power converters and drives: Advances and trends," *IEEE Trans. Ind. Electron.*, vol. 64, no. 2, pp. 935–947, Feb. 2017.

[55] J. F. Hu, Y. H. Shan, J. M. Guerrero, A. Ioinovici, K. W. Chan, and J. Rodriguez, "Model predictive control of microgrids — An overview," *Renew. Sustain. Energy Rev.*, vol. 136, p. 110422, Feb. 2021.

[56] A. Andersson and T. Thiringer, "Assessment of an improved finite control set model predictive current controller for automotive propulsion applications," *IEEE Trans. Ind. Electron.*, vol. 67, no. 1, pp. 91–100, Jan. 2020.

[57] W. H. Chen, J. Yang, L. Guo, and S. H. Li, "Disturbance-observer-based control and related methods — An overview," *IEEE Trans. Ind. Electron.*, vol. 63, no. 2, pp. 1083–1095, Feb. 2016.

[58] A. Damiano, G. Gatto, I. Marongiu, A. Perfetto, and A. Serpi, "Operating constraints management of a surface-mounted PM synchronous machine by means of an FPGA-based model predictive control algorithm," *IEEE Trans. Ind. Inform.*, vol. 10, no. 1, pp. 243–255, Feb. 2014.

[59] T. Zanma, S. Tozawa, Y. Takagi, K. Koiwa, and K. Z. Liu, "Optimal voltage vector in current control of PMSM considering torque ripple and reduction of the number of switching operations," *IET Power Electron.*, vol. 13, no. 6, pp. 1200–1206, May 2020.

[60] H. T. Nguyen and J.-W. Jung, "Asymptotic stability constraints for direct horizon-one model predictive control of SPMSM drives," *IEEE Trans. Power Electron.*, vol. 33, no. 10, pp. 8213–8219, Oct. 2018.

[61] S. Lin, X. Fang, X. Wang, Z. Yang and F. Lin, "Multiobjective model predictive current control method of permanent magnet synchronous traction motors with multiple current bounds in railway application," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 12348–12357, Dec. 2022.

[62] J. Yang, W. H. Chen, S. H. Li, L. Guo, and Y. D. Yan, "Disturbance/uncertainty estimation and attenuation techniques in PMSM drives — A survey," *IEEE Trans. Ind. Electron.*, vol. 64, no. 4, pp. 3273–3285, Apr. 2017.

[63] X. D. Sun, Y. Zhang, Y. F. Cai, and X. Tian, "Compensated deadbeat predictive current control considering disturbance and VSI nonlinearity for in-wheel PMSMs," *IEEE/ASME Trans. Mechatron.*, 2021, DOI: 10.1109/TMECH.2021.3135936.

[64] Y. A. R. I. Mohamed, "Design and implementation of a robust current-control scheme for a PMSM vector drive with a simple adaptive disturbance observer," *IEEE Trans. Ind. Electron.*, vol. 54, no. 4, pp. 1981–1988, Aug. 2007.

[65] M. Abdelrahem, C. M. Hackl, Z. B. Zhang, and R. Kennel, "Robust predictive control for direct-driven surface-mounted permanent-magnet synchronous generators without mechanical sensors," *IEEE Trans. Energy Convers.*, vol. 33, no. 1, pp. 179–189, Mar. 2018.

[66] X. D. Sun, Y. Zhang, G. Lei, Y. G. Guo, and J. G. Zhu, "An improved deadbeat predictive stator flux control with reduced-order disturbance observer for in-wheel PMSMs," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 2, pp. 690–700, Apr. 2022.

[67] C. F. Zhang, G. Wu, F. Rong, J. H. Feng, L. H. Jia, J. He, and S. D. Huang, "Robust fault-tolerant predictive current control for permanent magnet synchronous motors considering demagnetization fault," *IEEE Trans. Ind. Electron.*, vol. 65, no. 7, pp. 5324–5334, Jul. 2018.

[68] G. J. Pei, L. Y. Li, X. N. Gao, J. X. Liu, and R. Kennel, "Predictive current trajectory control for PMSM at voltage limit," *IEEE Access*, vol. 8, pp. 1670–1679, Jan. 2020.

[69] T. X. Li, R. Q. Ma, and W. J. Han, "Virtual-vector-based model predictive current control of five-phase PMSM with stator current and concentrated disturbance observer," *IEEE Access*, vol. 8, pp. 212635–212646, Dec. 2020.

[70] S. X. Niu, Y. X. Luo, W. N. Fu, and X. D. Zhang, "Robust model predictive control for a three-phase PMSM motor with improved control precision," *IEEE Trans. Ind. Electron.*, vol. 68, no. 1, pp. 838–849, Jan. 2021.

[71] F. Mwasilu, H. T. Nguyen, H. H. Choi, and J. W. Jung, "Finite set model predictive control of interior PM synchronous motor drives with an external disturbance rejection technique," *IEEE/ASME Trans. Mechatron.*, vol. 22, no. 2, pp. 762–773, Apr. 2017.

[72] X. G. Zhang, L. Zhang, and Y. C. Zhang, "Model predictive current control for PMSM drives with parameter robustness improvement," *IEEE Trans. Power Electron.*, vol. 34, no. 2, pp. 1645–1657, Feb. 2019.

[73] X. G. Zhang, Y. Cheng, and L. Zhang, "Disturbance-deadbeat inductance observer-based current predictive control for surface-mounted permanent magnet synchronous motors drives," *IET Power Electron.*, vol. 13, no. 6, pp. 1172–1180, May 2020.

[74] C. K. Lin, T. H. Liu, J. T. Yu, L. C. Fu, and C. F. Hsiao, "Model-free predictive current control for interior permanent-magnet synchronous motor drives based on current difference detection technique," *IEEE Trans. Ind. Electron.*, vol. 61, no. 2, pp. 667–681, Feb. 2014.

[75] M. S. Mubarok and T. H. Liu, "Implementation of predictive controllers for matrix-converter-based interior permanent magnet synchronous motor position control systems," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 7, no. 1, pp. 261–273, Mar. 2019.

[76] C. K. Lin, J. T. Yu, Y. S. Lai, H. C. Yu, Y. H. Lin, and F. M. Chen, "Simplified model-free predictive current control for interior permanent magnet synchronous motors," *Electron. Lett.*, vol. 52, no. 1, pp. 49–50, 2016.

[77] C. K. Lin, J. T. Yu, Y. S. Lai, and H. C. Yu, "Improved model-free predictive current control for synchronous reluctance motor drives," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3942–3953, Jun. 2016.

[78] C. A. Agustin, J. T. Yu, Y. S. Cheng, C. K. Lin, and Y. W. Yi, "A synchronized current difference updating technique for model-free predictive current control of PMSM drives," *IEEE Access*, vol. 9, pp. 63306–63318, May 2021.

[79] X. Li, Y. Wang, X. Guo, X. Cui, S. Zhang, and Y. Li, "An improved model-free current predictive control method for SPMSM drives," *IEEE Access*, vol. 9, pp. 134672–134681, 2021.

[80] X. Yuan, S. Zhang, C. N. Zhang, A. Galassini, G. Buticchi, and M. Degano, "Improved model predictive current control for SPMSM drives using current update mechanism," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 1938–1948, Mar. 2021.

[81] X. Yuan, S. Zhang, and C. N. Zhang, "Nonparametric predictive

current control for PMSM," *IEEE Trans. Power Electron.*, vol. 35, no. 9, pp. 9332–9341, Sept. 2020.

[82] Z. Y. Chen, J. Q. Qiu, and M. J. Jin, "Adaptive finite-control-set model predictive current control for IPMSM drives with inductance variation," *IET Electr. Power Appl.*, vol. 11, no. 5, pp. 874–884, May 2017.

[83] S. Wang, R. J. Zhao, W. M. Chen, G. D. Li, and C. Liu, "Parameter identification of PMSM based on windowed least square algorithm," *Adv. Mater. Res.*, vol. 383-390, pp. 5940–5944, Nov. 2011.

[84] N. Urasaki, Y. Noguchi, A. M. Howlader, Y. Yonaha, A. Yona, and T. Senjyu, "Wide-speed range operation of interior permanent magnet synchronous motor with parameter identification," *Electric Power Compon. Syst.*, vol. 37, no. 8, pp. 847–865, Jun. 2009.

[85] Y. C. Shi, K. Sun, L. Huang, and Y. D. Li, "Online identification of permanent magnet flux based on extended kalman filter for IPMSM drive with position sensorless control," *IEEE Trans. Ind. Electron.*, vol. 59, no. 11, pp. 4169–4178, Nov. 2012.

[86] C. Q. Zhong and Y. Y. Lin, "Model reference adaptive control (MRAC)-based parameter identification applied to surface-mounted permanent magnet synchronous motor," *Int. J. Electron.*, vol. 104, no. 11, pp. 1854–1873, Jun. 2017.

[87] Y. Chen, F. Zhou, X. Liu, and E. Hu, "Online adaptive parameter identification of PMSM based on the dead-time compensation," *Int. J. Electron.*, vol. 102, no. 7, pp. 1132–1150, Jul. 2015.

[88] Z. Q. Wang, M. B. Yang, L. Gao, Z. X. Wang, G. Z. Zhang, H. M. Wang, and X. Gu, "Deadbeat predictive current control of permanent magnet synchronous motor based on variable step-size adaline neural network parameter identification," *IET Electr. Power Appl.*, vol. 14, no. 11, pp. 2007–2015, Nov. 2020.

[89] O. Wallscheid and J. Böcker, "Global identification of a low-order lumped-parameter thermal network for permanent magnet synchronous motors," *IEEE Trans. Energy Convers.*, vol. 31, no. 1, pp. 354–365, Mar. 2016.

[90] G. Gatto, I. Marongiu, and A. Serpi, "Discrete-time parameter identification of a surface-mounted permanent magnet synchronous machine," *IEEE Trans. Ind. Electron.*, vol. 60, no. 11, pp. 4869–4880, Nov. 2013.

[91] J. Sawma, F. Khatounian, E. Monmasson, L. Idkhajine, and R. Ghosn, "Analysis of the impact of online identification on model predictive current control applied to permanent magnet synchronous motors," *IET Electr. Power Appl.*, vol. 11, no. 5, pp. 864–873, May 2017.

[92] L. H. Wang, G. J. Tan, and J. Meng, "Research on model predictive control of IPMSM based on adaline neural network parameter identification," *Energies*, vol. 12, no. 24, p. 4803, Dec. 2019.

[93] A. Brosch, S. Hanke, O. Wallscheid, and J. Böcker, "Data-driven recursive least squares estimation for model predictive current control of permanent magnet synchronous motors," *IEEE Trans. Power Electron.*, vol. 36, no. 2, pp. 2179–2190, Feb. 2021.

[94] H. Ahn, H. Park, C. Kim, and H. Lee, "A review of state-of-the-art techniques for PMSM parameter identification," *J. Electr. Eng. Technol.*, vol. 15, no. 3, pp. 1177–1187, Mar. 2020.

[95] M. S. Rafaq and J. W. Jung, "A comprehensive review of state-of-the-art parameter estimation techniques for permanent magnet synchronous motors in wide speed range," *IEEE Trans. Ind. Inf.*, vol. 16, no. 7, pp. 4747–4758, Jul. 2020.

[96] M. Preindl and S. Bolognani, "Model predictive direct torque control with finite control set for PMSM drive systems, part 2: Field weakening operation," *IEEE Trans. Ind. Inf.*, vol. 9, no. 2, pp. 648–657, May 2013.

[97] L. Chen, H. Xu, and X. D. Sun, "A novel strategy of control performance improvement for six-phase permanent magnet synchronous hub motor drives of EVs under new european driving cycle," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5628–5637, Jun. 2021.

[98] M. Uddin, S. Mekhilef, M. Mubin, M. Rivera, and J. Rodriguez, "Model predictive torque ripple reduction with weighting factor optimization fed by an indirect matrix converter," *Electr. Power Compon. Syst.*, vol. 42, no. 10, pp. 1059–1069, Jun. 2014.

[99] Z. Song, F. Zhou, Y. Yu, R. Zhang, and S. Hu, "Open-phase fault-tolerant predictive control strategy for open-end-winding permanent magnet synchronous machines without postfault controller reconfiguration," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 3770–3781, May 2021.

[100] X. G. Wang, W. Xu, Y. Zhao, and X. H. Li, "Modified MPC algorithm for NPC inverter fed disc coreless permanent magnet synchronous motor," *IEEE Trans. Appl. Supercond.*, vol. 26, no. 7, p. 0608505, Oct. 2016.

[101] W. S. Wang, Y. Fan, S. Y. Chen, and Q. S. Zhang, "Finite control set model predictive current control of a five-phase PMSM with virtual voltage vectors and adaptive control set," *CES Trans. Electr. Mach. Syst.*, vol. 2, no. 1, pp. 136–141, Mar. 2018.

[102] Y. Yan, S. Wang, C. L. Xia, H. M. Wang, and T. N. Shi, "Hybrid control set-model predictive control for field-oriented control of VSI-PMSM," *IEEE Trans. Energy Convers.*, vol. 31, no. 4, pp. 1622–1633, Dec. 2016.

[103] Y. X. Luo and C. H. Liu, "A flux constrained predictive control for a six-phase PMSM motor with lower complexity," *IEEE Trans. Ind. Electron.*, vol. 66, no. 7, pp. 5081–5093, Jul. 2019.

[104] X. Gu, Sh en, X. M. Li, G. Z. Zhang, Z. Q. Wang, and T. N. Shi, "Improved vector selection based model predictive torque control for IPMSM," *IET Electr. Power Appl.*, vol. 14, no. 1, pp. 139–146, Jan. 2020.

[105] P. Kakosimos, S. Bayhan, and H. Abu-Rub, "Predictive control with uniform switching transitions and reduced calculation requirements," in *Proc. 43rd Annu. Conf. IEEE Industrial Electronics Society*, Beijing, China, 2017, pp. 6342–6347.

[106] W. Xie, X. C. Wang, F. X. Wang, W. Xu, R. M. Kennel, D. Gerling, and R. D. Lorenz, "Finite-control-set model predictive torque control with a deadbeat solution for PMSM drives," *IEEE Trans. Ind. Electron.*, vol. 62, no. 9, pp. 5402–5410, Sept. 2015.

[107] Y. C. Zhang, D. L. Xu, J. L. Liu, S. Y. Gao, and W. Xu, "Performance improvement of model-predictive current control of permanent magnet synchronous motor drives," *IEEE Trans. Ind. Appl.*, vol. 53, no. 4, pp. 3683–3695, Jul.–Aug. 2017.

[108] Y. H. Li, Y. G. Qin, Y. F. Zhou, and C. H. Zhao, "Model predictive torque control for permanent magnet synchronous motor based on dynamic finite-control-set using fuzzy control," *Energy Rep.*, vol. 6, no. S9, pp. 128–133, Dec. 2020.

[109] J. Rodriguez, J. Pontt, C. A. Silva, Correa, Lezana, Cortes, and U. Ammann, "Predictive current control of a voltage source inverter," *IEEE Trans. Ind. Electron.*, vol. 54, no. 1, pp. 495–503, Feb. 2007.

[110] Cortés, J. Rodríguez, Antoniewicz, and M. Kazmierkowski, "Direct power control of an AFE using predictive control," *IEEE Trans. Power Electron.*, vol. 23, no. 5, pp. 2516–2523, Sept. 2008.

[111] R. O. Ramírez, C. R. Baier, F. Villarroel, J. R. Espinoza, J. Pou, and J. Rodríguez, "A hybrid FCS-MPC with low and fixed switching frequency without steady-state error applied to a grid-connected CHB inverter," *IEEE Access*, vol. 8, pp. 223637–223651, Dec. 2020.

[112] R. O. Ramírez, C. R. Baier, J. Espinoza, and F. Villarroel, "Finite control set MPC with fixed switching frequency applied to a grid connected single-phase cascade h-bridge inverter," *Energies*, vol. 13, no. 20, p. 5475, Oct. 2020.

[113] X. Zhang, G. J. Tan, T. Xia, Q. Wang, and X. Wu, "Optimized switching finite control set model predictive control of NPC single-phase three-level rectifiers," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 10097–10108, Oct. 2020.

[114] C. Xiong, H. Xu, T. Guan, and Zhou, "A Constant switching frequency multiple-vector-based model predictive current control of five-phase PMSM with nonsinusoidal back EMF," *IEEE Trans. Ind. Electron.*, vol. 67, no. 3, pp. 1695–1707, Mar. 2020.

[115] F. Zhang, T. Peng, H. B. Dan, J. H. Lin, and M. Su, "Modulated model predictive control of permanent magnet synchronous motor," in *Proc. IEEE Int. Conf. Industrial Electronics for Sustainable Energy Systems*, Hamilton, New Zealand, 2018, pp. 130–133.

[116] S. S. Yeoh, T. Yang, L. Tarisciotti, C. I. Hill, S. Bozhko, and Zanchetta, "Permanent-magnet machine-based starter–generator system with modulated model predictive control," *IEEE Trans. Transp. Electrific.*, vol. 3, no. 4, pp. 878–890, Dec. 2017.

[117] C. F. Garcia, C. A. Silva, J. R. Rodriguez, Zanchetta, and S. A. Odhano, "Modulated model-predictive control with optimized overmodulation," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 7, no. 1, pp. 404–413, Mar. 2019.

[118] Q. Wang, H. T. Yu, C. Li, X. Y. Lang, S. S. Yeoh, T. Yang, M.

Rivera, S. Bozhko, and Wheeler, "A low-complexity optimal switching time-modulated model-predictive control for PMSM with three-level NPC converter," *IEEE Trans. Transp. Electrific.*, vol. 6, no. 3, pp. 1188–1198, Sept. 2020.

[119] S. G. Petkar, K. Eshwar, and V. K. Thippiripati, "A modified model predictive current control of permanent magnet synchronous motor drive," *IEEE Trans. Ind. Electron.*, vol. 68, no. 2, pp. 1025–1034, Feb. 2021.

[120] W. Chen, S. K. Zeng, G. Z. Zhang, T. N. Shi, and C. L. Xia, "A modified double vectors model predictive torque control of permanent magnet synchronous motor," *IEEE Trans. Power Electron.*, vol. 34, no. 11, pp. 11419–11428, Nov. 2019.

[121] X. G. Zhang and B. S. Hou, "Double vectors model predictive torque control without weighting factor based on voltage tracking error," *IEEE Trans. Power Electron.*, vol. 33, no. 3, pp. 2368–2380, Mar. 2018.

[122] Y. Xu, X. H. Ding, J. B. Wang, and C. Wang, "Robust three-vector-based low-complexity model predictive current control with supertwisting-algorithm-based second-order sliding-mode observer for permanent magnet synchronous motor," *IET Power Electron.*, vol. 12, no. 11, pp. 2895–2903, Sept. 2019.

[123] L. Chen, H. Xu, X. D. Sun, and Y. F. Cai, "Three-vector-based model predictive torque control for a permanent magnet synchronous motor of EVs," *IEEE Trans. Transp. Electrific.*, vol. 7, no. 3, pp. 1454–1465, Sept. 2021.

[124] Y. Xu, X. H. Ding, J. B. Wang, and Y. Y. Li, "Three-vector-based low-complexity model predictive current control with reduced steady-state current error for permanent magnet synchronous motor," *IET Electr. Power Appl.*, vol. 14, no. 2, pp. 305–315, Feb. 2020.

[125] S. W. Kang, J. H. Soh, and R. Y. Kim, "Symmetrical three-vector-based model predictive control with deadbeat solution for IPMSM in rotating reference frame," *IEEE Trans. Ind. Electron.*, vol. 67, no. 1, pp. 159–168, Jan. 2020.

[126] Cortes, J. Rodriguez, C. Silva, and A. Flores, "Delay compensation in model predictive current control of a three-phase inverter," *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 1323–1325, Feb. 2012.

[127] Y. Yang, H. Q. Wen, and D. P. Li, "A fast and fixed switching frequency model predictive control with delay compensation for three-phase inverters," *IEEE Access*, vol. 5, pp. 17904–17913, Sept. 2017.

[128] T. Jin, X. Y. Shen, T. X. Su, and R. C. C. Flesch, "Model predictive voltage control based on finite control set with computation time delay compensation for PV systems," *IEEE Trans. Energy Convers.*, vol. 34, no. 1, pp. 330–338, Mar. 2019.

[129] X. D. Sun, J. H. Cao, G. Lei, Y. G. Guo, and J. G. Zhu, "A robust deadbeat predictive controller with delay compensation based on composite sliding mode observer for PMSMs," *IEEE Trans. Power Electron.*, vol. 36, no. 9, pp. 10742–10752, Sept. 2021.

[130] Y. X. Luo and C. H. Liu, "Model predictive control for a six-phase PMSM motor with a reduced-dimension cost function," *IEEE Trans. Ind. Electron.*, vol. 67, no. 2, pp. 969–979, Feb. 2020.

[131] Y. F. Han, C. Gong, L. M. Yan, H. Q. Wen, Y. G. Wang, and K. Shen, "Multiobjective finite control set model predictive control using novel delay compensation technique for PMSM," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 11193–11204, Oct. 2020.

[132] F. X. Wang, J. X. Wang, R. M. Kennel, and J. Rodríguez, "Fast speed control of AC machines without the proportional-integral controller: Using an extended high-gain state observer," *IEEE Trans. Power Electron.*, vol. 34, no. 9, pp. 9006–9015, Sept. 2019.

[133] F. X. Wang, H. T. Xie, Q. Chen, S. A. Davari, J. Rodríguez, and R. Kennel, "Parallel predictive torque control for induction machines without weighting factors," *IEEE Trans. Power Electron.*, vol. 35, no. 2, pp. 1779–1788, Feb. 2020.

[134] P. Cortes, S. Kouro, B. La Rocca, R. Vargas, J. Rodriguez, J. I. Leon, S. Vazquez, and L. G. Franquelo, "Guidelines for weighting factors design in model predictive control of power converters and drives," in *Proc. IEEE Int. Conf. Industrial Technology*, Churchill, Australia, 2009, pp. 1–7.

[135] R. U. Guazzelli, W. C. De Andrade Pereira, C. M. R. De Oliveira, A. G. De Castro, and M. L. De Aguiar, "Weighting factors optimization of predictive torque control of induction motor by multiobjective genetic algorithm," *IEEE Trans. Power Electron.*, vol. 34, no. 7, pp. 6628–6638, Jul. 2019.

[136] M. K. Wu, X. D. Sun, J. G. Zhu, G. Lei, and Y. G. Guo, "Improved model predictive torque control for PMSM drives based on duty cycle optimization," *IEEE Trans. Magn.*, vol. 57, no. 2, pp. 1–5, Feb. 2021.

[137] Y. X. Luo and C. H. Liu, "Elimination of harmonic currents using a reference voltage vector based-model predictive control for a six-phase PMSM motor," *IEEE Trans. Power Electron.*, vol. 34, no. 7, pp. 6960–6972, Jul. 2019.

[138] X. G. Zhang and Y. K. He, "Direct voltage-selection based model predictive direct speed control for PMSM drives without weighting factor," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7838–7851, Aug. 2019.

[139] G. Z. Zhang, C. Chen, X. Gu, Z. Q. Wang, and X. M. Li, "An improved model predictive torque control for a two-level inverter fed interior permanent magnet synchronous motor," *Electronics*, vol. 8, no. 7, p. 769, Jul. 2019.

[140] D. Sun, J. Su, C. Sun, and H. Nian, "A simplified MPFC with capacitor voltage offset suppression for the four-switch three-phase inverter-fed PMSM drive," *IEEE Trans. Ind. Electron.*, vol. 66, no. 10, pp. 7633–7642, Oct. 2019.

[141] X. S. Wu, W. S. Song, and C. Xue, "Low-complexity model predictive torque control method without weighting factor for five-phase PMSM based on hysteresis comparators," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 6, no. 4, pp. 1650–1661, Dec. 2018.

[142] T. Geyer and D. E. Quevedo, "Multistep finite control set model predictive control for power electronics," *IEEE Trans. Power Electron.*, vol. 29, no. 12, pp. 6836–6846, Dec. 2014.

[143] T. Geyer and D. E. Quevedo, "Performance of multistep finite control set model predictive control for power electronics," *IEEE Trans. Power Electron.*, vol. 30, no. 3, pp. 1633–1644, Mar. 2015.

[144] Karamanakos, T. Geyer, and R. Aguilera, "Long-horizon direct model predictive control: Modified sphere decoding for transient operation," *IEEE Trans. Ind. Appl.*, vol. 54, no. 6, pp. 6060–6070, Nov.–Dec. 2018.

[145] R. Baidya, R. P. Aguilera, P. Acuña, T. Geyer, R. A. Delgado, D. E. Quevedo, and H. Du Toit Mouton, "Enabling multistep model predictive control for transient operation of power converters," *IEEE Open J. Ind. Electron. Soc.*, vol. 1, pp. 284–297, Oct. 2020.

[146] J. F. Hu, J. G. Zhu, G. Lei, G. Platt, and D. G. Dorrell, "Multi-objective model-predictive control for high-power converters," *IEEE Trans. Energy Convers.*, vol. 28, no. 3, pp. 652–663, Sept. 2013.

[147] Y. J. Yu and X. Z. Wang, "Multi-step predictive current control for NPC grid-connected inverter," *IEEE Access*, vol. 7, pp. 157756–157765, Oct. 2019.

[148] W. Chen, X. H. Zhang, X. Gu, Y. Yan, and T. N. Shi, "Band-based multi-step predictive torque control strategy for PMSM drives," *IEEE Access*, vol. 7, pp. 171411–171422, Nov. 2019.

[149] Y. L. Wang, W. Xie, X. C. Wang, W. B. Yang, M. F. Dou, S. J. Song, and D. Gerling, "Fast response model predictive torque and flux control with low calculation effort for PMSMs," *IEEE Trans. Ind. Inform.*, vol. 15, no. 10, pp. 5531–5540, Oct. 2019.

[150] T. Dorfling, H. Du Toit Mouton, T. Geyer, and Karamanakos, "Long-horizon finite-control-set model predictive control with nonrecursive sphere decoding on an FPGA," *IEEE Trans. Power Electron.*, vol. 35, no. 7, pp. 7520–7531, Jul. 2020.

[151] C. A. Agustin, J. T. Yu, C. K. Lin, J. Jai, and Y. S. Lai, "Triple-voltage-vector model-free predictive current control for four-switch three-phase inverter-fed SPMSM based on discrete-space-vector modulation," *IEEE Access*, vol. 9, pp. 60352–60363, Apr. 2021.

[152] M. Yang, X. Y. Lang, J. Long, and D. G. Xu, "Flux Immunity robust predictive current control with incremental model and extended state observer for PMSM drive," *IEEE Trans. Power Electron.*, vol. 32, no. 12, pp. 9267–9279, Dec. 2017.

[153] M. Khalilzadeh and S. Vaez-Zadeh, "A robust predictive torque and flux control for IPM motor drives without a cost function," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 8067–8075, Jul. 2021.

[154] F. X. Wang, K. K. Zuo, P. Tao, and J. Rodríguez, "High performance model predictive control for PMSM by using stator current mathematical model self-regulation technique," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 13652–13662, Dec. 2020.

[155] K. S. Alam, M. Akter, D. Xiao, D. M. Zhang, and M. F. Rahman, "Asymptotically stable predictive control of grid-connected converter based on discrete space vector modulation," *IEEE Trans. Ind. Inf.*,

vol. 15, no. 5, pp. 2775–2785, May 2019.

[156] R. Aguilera and D. E. Quevedo, "Predictive control of power converters: Designs with guaranteed performance," *IEEE Trans. Ind. Inf.*, vol. 11, no. 1, pp. 53–63, Feb. 2015.

[157] H. T. Nguyen and J. W. Jung, "Finite control set model predictive control to guarantee stability and robustness for surface-mounted PM synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 65, no. 11, pp. 8510–8519, Nov. 2018.

[158] A. Katkout, T. Nasser, and A. Essadki, "Novel predictive control for the IPMSM fed by the 3L-SNPC inverter for EVAs: Modified Lyapunov function, computational efficiency, and delay compensation," *Math. Probl. Eng.*, vol. 2020, p. 2515107, Aug. 2020.

[159] G. Q. Bao, W. G. Qi, and T. He, "Direct torque control of PMSM with modified finite set model predictive control," *Energies*, vol. 13, no. 1, pp. 234, Jan. 2020.

[160] X. D. Sun, T. Li, X. Tian, and J. G. Zhu, "Fault-tolerant operation of a six-phase permanent magnet synchronous hub motor based on model predictive current control with virtual voltage vectors," *IEEE Trans. Energy Convers.*, vol. 37, no. 1, pp. 337–346, Mar. 2022.

[161] X. D. Sun, T. Li, Z. Zhu, G. Lei, Y. G. Guo, and J. G. Zhu, "Speed sensorless model predictive current control based on finite position set for PMSHM drives," *IEEE Trans. Transp. Electrific.*, vol. 7, no. 4, pp. 2743–2752, Dec. 2021.

[162] T. Yang, T. Kawaguchi, S. Hashimoto, and W. Jiang, "Switching sequence model predictive direct torque control of IPMSMs for EVs in switch open-circuit fault-tolerant mode," *Energies*, vol. 13, no. 21, p. 5593, Oct. 2020.

[163] Y. X. Luo and C. H. Liu, "Pre- and post-fault tolerant operation of a six-phase PMSM motor using FCS-MPC without controller reconfiguration," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 254–263, Jan. 2019.

[164] G. Forstner, A. Kugi, and W. Kemmetmüller, "Fault-tolerant torque control of a three-phase permanent magnet synchronous motor with inter-turn winding short circuit," *Control Eng. Pract.*, vol. 113, p. 104846, Aug. 2021.

[165] M. Abdelrahem, C. M. Hackl, J. Rodríguez, and R. Kennel, "Model reference adaptive system with finite-set for encoderless control of PMSGS in micro-grid systems," *Energies*, vol. 13, no. 18, p. 4844, Sept. 2020.

[166] L. Sun, X. X. Li, and L. M. Chen, "Motor speed control with convex optimization-based position estimation in the current loop," *IEEE Trans. Power Electron.*, vol. 36, no. 9, pp. 10906–10919, Sept. 2021.

[167] F. Liu, H. T. Li, L. Liu, R. M. Zou, and K. Z. Liu, "A control method for IPMSM based on active disturbance rejection control and model predictive control," *Mathematics*, vol. 9, no. 7, p. 760, Apr. 2021.

[168] H. J. Shi and X. C. Nie, "Composite control for disturbed direct-driven surface-mounted permanent magnet synchronous generator with model prediction strategy," *Meas. Control*, vol. 54, no. 5–6, pp. 1015–1025, Apr. 2021.

[169] Z. Y. Sun, S. Xu, G. Z. Ren, C. X. Yao, and G. T. Ma, "A cascaded band based model predictive current control for PMSM drives," *IEEE Trans. Ind. Electron.*, 2022, DOI: 10.1109/TIE.2022.3176312.

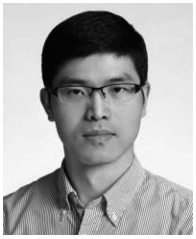
[170] J. Yoo and K. H. Johansson, "Event-triggered model predictive control with a statistical learning," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 51, no. 4, pp. 2571–2581, Apr. 2021.

[171] D. W. Shi, J. Xue, L. X. Zhao, J. Z. Wang, and Y. Huang, "Event-triggered active disturbance rejection control of DC torque motors," *IEEE/ASME Trans. Mechatron.*, vol. 22, no. 5, pp. 2277–2287, Oct. 2017.



Teng Li received the B.S. degree in vehicle engineering from Jiangsu University, in 2019, and he is currently working toward the Ph.D. degree at Jiangsu University.

His current research interests include modelling, structure designing and controlling of permanent magnet synchronous motors for electric vehicle propulsion.



Xiaodong Sun (Senior Member, IEEE) received the B.Sc. degree in electrical engineering, and the M.Sc. and Ph.D. degrees in control engineering from Jiangsu University, in 2004, 2008, and 2011, respectively.

Since 2004, he has been with Jiangsu University, where he is currently a Professor in vehicle engineering with the Automotive Engineering Research Institute. From 2014 to 2015, he was a Visiting Professor with the School of Electrical, Mechanical, and Mechatronic Systems, University of Technology Sydney, Australia. His current teaching and research interests include electrified vehicles, electrical machines, electrical drives, and energy management. He is the author or coauthor of more than 100 refereed technical papers and one book, and he is the holder of 42 patents in his areas of interest.

Dr. Sun is an Associate Editor of the *IEEE Trans. Ind. Electron*. and an Editor of the *IEEE Trans. Energy Convers*.



Gang Lei (Member, IEEE) received the B.S. degree in mathematics from Huanggang Normal University, in 2003, the M.S. degree in mathematics and Ph.D. degree in electrical engineering from Huazhong University of Science and Technology, in 2006 and 2009, respectively.

He is currently a Senior Lecturer at the School of Electrical and Data Engineering, University of Technology Sydney (UTS), Australia. His research interests include computational electromagnetics, design optimization and control of electrical drive systems and renewable energy systems. He is an Associate Editor of the *IEEE Trans. Ind. Electron.*.



Zebin Yang received the B.Sc., M.Sc. and Ph.D. degrees in electrical engineering from Jiangsu University, in 1999, 2004, and 2013, respectively, where he is currently a Professor at the School of Electrical Information Engineering, Jiangsu University. From 2014 to 2015, he was a Visiting Professor with the School of Electrical, Mechanical, and Mechatronic Systems, University of Technology Sydney, Australia. His current teaching and research interests include electrical machines and electrical drives, bearingless motors and magnetic levitation transmission technology.



Youguang Guo (Senior Member, IEEE) received the B.E. degree from Huazhong University of Science and Technology in 1985, the M.E. degree from Zhejiang University in 1988, and the Ph.D. degree from University of Technology, Australia in 2004, all in electrical engineering.

He is currently a Professor at the School of Electrical and Data Engineering, University of Technology Sydney (UTS), Australia. His research interests include measurement and modeling of properties of magnetic materials, numerical analysis of electromagnetic field, electrical machine design optimization, and power electronic drives and control.



Jianguo Zhu (Senior Member, IEEE) received the B.E. degree in 1982 from Jiangsu Institute of Technology, the M.E. degree in 1987 from Shanghai University of Technology, and the Ph.D. degree in 1995 from the University of Technology Sydney, Australia, all in electrical engineering.

He was appointed a Lecturer at UTS in 1994 and promoted to Full Professor in 2004 and Distinguished Professor of Electrical Engineering in 2017. At UTS, he has held various leadership positions, including the Head of School for School of Electrical, Mechanical and Mechatronic Systems and Director for Centre of Electrical Machines and Power Electronics. In 2018, he joined the University of Sydney, Australia, as a Full Professor and Head of the School for School of Electrical and Information Engineering. His research interests include computational electromagnetics, measurement and modelling of magnetic properties of materials, electrical machines and drives, power electronics, renewable energy systems, and smart micro grids.